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Different forms of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide

Abstract

The present invention relates to different forms of the compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide and to methods of making such forms/compounds. The present invention furthermore relates to mono-acid addition salts thereof, to methods of making such mono-acid addition salts and to pharmaceutical compositions comprising any of the aforementioned compounds. Furthermore, the present invention relates to uses of any of these compounds.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage Application of International Application Number PCT/EP2019/070524, filed Jul. 30, 2019.

FIELD OF THE INVENTION

(2) The present invention relates to different forms of the compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide and to methods of making such forms/compounds. The present invention furthermore relates to mono-acid addition salts thereof, to methods of making such mono-acid addition salts and to pharmaceutical compositions comprising any of the aforementioned compounds. Furthermore, the present invention relates to uses of any of these compounds.

BACKGROUND OF THE INVENTION

(3) Tuberculosis as a disease continues to result in millions of deaths each year. Inadequate use of chemotherapy has led to an increasing number of drug resistant cases. This situation is likely to worsen with the emergence of extremely resistant strains to all currently known drugs. Current chemotherapy consists of compounds that directly target *Mycobacterium tuberculosis*, either by neutralizing general information pathways and critical processes such as RNA polymerization and protein synthesis inhibition or by interfering with mycobacterial specific cell envelop synthesis. The most widely used dedicated anti-tubercular drugs isoniazid, ethionamide, and pyrazin amide are pro-drugs that first require activation. They are administered to a patient for a course of several months. Patients infected with multi-drug resistant strains of *M. tuberculosis* may have to undergo combination therapies for extended periods of time.

(4) WO 2011/113606 describes various anti-tubercular compounds and their use in the treatment of bacterial infections, including compound "Q203" which chemically is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide. In a publication by Pethe et al. (Nature Medicine, 19, 1157-1160 (2013), this compound is reported to be active against tuberculosis by interfering with the bacterial energy metabolism, inhibiting cytochrome bcl activity which is an essential component of the electron transport chain required for synthesis of ATP.

(5) Whilst the compound shows promise for future therapy of tuberculosis and related infections, there continues to be a need for forms thereof that are particularly suitable for pharmaceutical administration. In particular there is a need to provide forms that are showing an improved solubility in comparison to the free base of this compound. Furthermore, there is a need in the art to provide for forms that show an improved stability.

BRIEF SUMMARY

(6) In a first aspect the present invention relates to a compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide ditosylate having the structure

(7) ##STR00001## and furthermore having at least one or several of the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 3.9° 2θ, 5.6° 2θ, 8.0° 2θ, 16.1° 2θ, 19.1° 2θ, and 22.4° 2θ, ±0.2° 2θ.

(8) In one embodiment, the compound has an XRPD spectrum as shown in FIG. 1.

(9) In one embodiment, the compound has a differential scanning calorimetry (DSC) thermogram showing a single endotherm peak with an onset of 235° C.-237° C.

(10) In one embodiment, the compound is produced by a method comprising the steps: Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2; Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone; Evaporating the solvent or solvent mixture.

- (11) In a further aspect, the present invention relates to a method for making the compound as defined above, said method comprising the steps: Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2; Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone; Evaporating the solvent or solvent mixture.
- (12) In a further aspect, the present invention relates to a mono-acid addition salt of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide which is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-hydrochloride, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-phosphate, or 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-tosylate.
- (13) In one embodiment, the mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-hydrochloride having at least one or several of the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 6.4° 2θ, 8.1° 2θ, 16.2° 2θ, 17.2° 2θ, 24.3° 2θ and 25.0° 2θ, ±0.2° 2θ.
- (14) In one embodiment, the mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-phosphate having at least one or several of the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 9.0° 2θ, 10.7±0.2° 2θ, 11.7° 2θ, 14.8° 2θ, 18.4° 2θ, 19.3° 2θ, and 21.8° 2θ, 22.8° 2θ, ±0.2° 2θ.
- (15) In one embodiment, the mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-tosylate having at least one or several of the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 4.0° 2θ, 11.4° 2θ, 12.2° 2θ, 14.4° 2θ, 17.7° 2θ, 18.9° 2θ, 19.7° 2θ, 20.3° 2θ, 23.2° 2θ, and 26.7° 2θ, +0.2° 2θ.
- (16) In one embodiment, the mono-acid addition salt is the mono-hydrochloride salt and has an XRPD spectrum as shown in FIG. 2A.
- (17) In one embodiment, the mono-acid addition salt is the mono-phosphate salt and has an XRPD spectrum as shown in FIG. 2B.
- (18) In one embodiment, the mono-acid addition salt is the mono-tosylate salt and has an XRPD spectrum as shown in FIG. 3.
- (19) In a further aspect, the present invention relates to a method for preparing the mono-acid addition salt, as defined above, said method comprising the steps: Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and an acid selected from hydrochloric acid, phosphoric acid and para-toluenesulphonic acid in a stoichiometric ratio of 1:1; Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), methyl tert-butyl ether (MTBE), tetrahydrofuran (THF), acetone or a mixture of THF and acetone; Evaporating the solvent or solvent mixture.
- (20) In a further aspect, the present invention relates to a pharmaceutical composition comprising at least one compound according to the present invention or a mono-acid addition salt according to the present invention, together with at least one pharmaceutically acceptable carrier, excipient and/or diluent.
- (21) In one embodiment, the pharmaceutical composition further comprises at least one other pharmaceutically active agent.
- (22) In a further aspect, the present invention relates to the compound or the mono-acid addition salt according to the present invention as defined above, for use in the treatment of a bacterial

infection.

(23) In one embodiment, said bacterial infection is tuberculosis or Buruli ulcer.

(24) In a further aspect, the present invention relates to a method treatment of a bacterial infection, in particular tuberculosis or Buruli ulcer, comprising the application of a suitable amount of a compound or of a mono-acid addition salt according to the present invention or of a pharmaceutical composition according to the present invention, to a patient in need thereof.

(25) The inventors have found that a particular form of the compound's ditosylate salt, a crystal polymorph form, sometimes herein also designated as "pattern A" or "form A" was particularly stable, and other forms converted into such stable polymorph form. In an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation, such form has at least one or several of the following peaks: 3.9° 2θ, 5.6° 2θ, 8.0° 2θ, 16.1° 2θ, 19.1° 2θ, and 22.4° 2θ, 0.2° 2θ.

(26) The 2θ-values have a standard deviation of ±0.2° 2θ. In one embodiment, the compound has an XRPD-spectrum as shown in FIG. 1.

(27) In comparison to other forms, this form appears to be the most stable form which makes it therefore particularly suitable for pharmaceutical formulations. In one embodiment, the compound according to the present invention has a differential scanning calorimetry (DSC) thermogram showing a single endotherm peak with an onset of approximately 235° C.-237° C.

(28) In one embodiment, the compound according to the present invention has a differential scanning calorimetry (DSC) thermogram showing a single endotherm peak with an onset of 235° C.-237° C.

(29) In one embodiment, the compound according to the present invention is produced by a method comprising the steps: Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2; Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone; Evaporating the solvent or solvent mixture.

(30) The present invention also relates to a method for making the compound as defined above, said method comprising the steps: Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2; Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone; Evaporating the solvent or solvent mixture.

(31) Furthermore, the inventors have also found that the compound according to the present invention, i. e. 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide ditosylate is stable under long-term storage conditions of 25° C. and 60% relative humidity for up to 60 months and under accelerated conditions of 40° C. and 75% relative humidity up to at least six months.

(32) In a further aspect, the present invention also relates to a mono-acid addition salt of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide, which is either mono-hydrochloride, mono-phosphate or mono-tosylate.

(33) When the mono-acid addition salt is mono-hydrochloride, then the compound, in one embodiment has at least one or several of the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 6.4° 2θ, 8.1° 2θ, 16.2° 2θ, 17.2° 2θ, 24.3° 2θ and 25.0° 2θ, with all the 2θ values having a standard deviation±0.2° 2θ.

(34) When the mono-acid addition salt is mono-phosphate, then the compound, in one embodiment has at least one or several of the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 9.0° 2θ, 10.7±0.2° 2θ, 11.7° 2θ, 14.8° 2θ, 18.4° 2θ, 19.3° 2θ, and 21.8° 2θ, 22.8° 2θ, ±0.2° 2θ.

(35) When the mono-acid addition salt is the mono-tosylate, in one embodiment, it has at least one

or several of the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 4.0° 2θ, 11.4° 2θ, 12.2° 2θ, 14.4° 2θ, 17.7° 2θ, 18.9° 2θ, 19.7° 2θ, 20.3° 2θ, 23.2° 2θ, and 26.7° 2θ, with all the 2θ values having a standard deviation of 0.2° 2θ.

(36) In one embodiment of the aforementioned mono-acid addition salts, the respective compound has all of the aforementioned respective peaks in a X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α).

(37) In one embodiment, the mono-acid addition salt is the mono-hydrochloride and has an XRPD spectrum as shown in FIG. 2A.

(38) In one embodiment, the mono-acid addition salt is the mono-phosphate and has an XRPD spectrum as shown in FIG. 2B.

(39) In another embodiment it is the mono-tosylate and has a XRPD-spectrum as shown in FIG. 3.

(40) In a further aspect, the present invention also relates to a method for preparing the mono-acid addition salt, as defined before, the method comprising the steps: Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and an acid selected from hydrochloric acid, phosphoric acid and para-toluenesulphonic acid in a stoichiometric ratio of 1:1; Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), methyl tert-butyl ether (MTBE), tetrahydrofuran (THF), acetone or a mixture of THF and acetone; Evaporating the solvent or solvent mixture.

(41) The inventors have surprisingly found that the aforementioned three mono-acid addition salts have a greater solubility at low pH-values, in particular around pH1. This is important insofar, as this drug is supposed to be orally taken and therefore has to pass the gastrointestinal tract. An increased solubility, in comparison to the free base implies a greater bioavailability.

(42) When reference is made herein to a scenario where a compound is described as having “an XRPD spectrum, as shown hereafter”, this is meant to refer to a situation, where a compound has peaks and signals in an XRPD spectrum at positions as shown in the respective XRPD spectrum referred to. The intensity shown of individual peaks does not need to necessarily be identical, as long as there is a peak or signal at the indicated position, within the typical range(s) of tolerance for such XRPD spectra.

(43) Furthermore, at low pH, especially around pH1, the mono-hydrochloride and the mono-phosphate salt have a better solubility in comparison to the di-tosylate salt.

(44) Furthermore, the present invention also relates to a pharmaceutical composition comprising a compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide ditosylate as defined above, or a mono-acid addition salt of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide, as defined above, together with at least one pharmaceutically acceptable carrier, excipient and/or diluent.

(45) In one embodiment, such pharmaceutical composition further comprises at least one other pharmaceutically active agent.

(46) In a further aspect, the present invention relates to the ditosylate compound, as defined above, or the mono-acid addition salt, as defined above, for use in the treatment of a bacterial infection.

(47) In one embodiment, the bacterial infection is tuberculosis or Buruli ulcer.

(48) In a further aspect, the present invention also relates to a method of treatment of a bacterial infection, in particular tuberculosis or Buruli ulcer, comprising the application of suitable amount of compound as defined above or of a mono-acid addition salt as defined above, or of a pharmaceutical composition, as defined above, to a patient in need thereof.

(49) In yet a further aspect, the present invention also relates to the use of a ditosylate compound, as defined above, or of a mono-acid addition salt, as defined above, or of a pharmaceutical composition, as defined above, for the manufacture of a medicament for the treatment of a bacterial

infection, wherein preferably, the bacterial infection is tuberculosis or Buruli ulcer.
(50) It should be noted that, as used herein, the compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide is also sometimes herein referred to as “Q203”.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The invention is now further described by the following figures wherein
- (2) FIG. 1 shows an XRPD spectrum of Form A (or “pattern A”) of Q203 ditosylate.
- (3) FIGS. 2A and 2B show an XRPD spectrum of the mono-HCl-form and mono-phosphate form of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide, respectively.
- (4) FIG. 3 shows an XRPD spectrum of the mono-tosylate form of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide.
- (5) FIG. 4 shows a DSC scan of form A of Q203 ditosylate.
- (6) FIG. 5 shows a TGA scan of form A of Q203 ditosylate.
- (7) FIG. 6 shows XRPD scans of the different obtained forms described further below.
- (8) FIG. 7 shows DSC scans of the different obtained forms described further below.
- (9) FIG. 8 shows a DVS scan of Form A.
- (10) FIG. 9 shows an XRPD scan of Form C.
- (11) FIG. 10 shows a DSC scan of Form C.
- (12) FIG. 11 shows a TGA scan of Form C.
- (13) FIG. 12 shows a DVS of Form C.
- (14) FIG. 13 shows an XRPD scan of Form G solids from slurry experiments.
- (15) FIG. 14 shows a DSC scan of pattern G solids from slurry experiments.
- (16) FIG. 15 shows a TGA scan of pattern G solids from slurry experiments.
- (17) FIG. 16 shows an XRPD scan of Form A solids from 100% RH experiment.
- (18) FIG. 17 shows an XRPD scan of Form A solids from 100% RH experiment.
- (19) FIG. 18 shows an XRPD scan of Form C solids from 100% RH experiment.
- (20) FIG. 19 shows DSC scans of Form A solids from 100% RH experiment.
- (21) FIG. 20 shows DSC scan of Form A solids from 100% RH experiment.
- (22) FIG. 21 shows DSC scans of Form C solids from 100% RH experiment.
- (23) FIG. 22 shows an XRPD pattern of Q203 free base, batch C12032302-J16001.
- (24) FIG. 23 shows XRPD pattern of Q203 ditosylate, batch C12032302-K16001M, form A (i. e. proper “Q203” ditosylate).
- (25) FIG. 24 shows XRPD patterns of Q203 free base (top trace), ditosylate salt (Pattern A, 2nd trace from top), ND-0006E-007-16 (3.sup.rd trace from top) and pTSA (bottom trace).
- (26) FIG. 25 shows XRPD patterns of Q203 free base (top trace) and ND-0006E-003-01 (Type 2, bottom trace).
- (27) FIG. 26 shows XRPD patterns of Q203 free base (top trace) and ND-0006E-003-25 (Type 3, bottom trace).
- (28) FIG. 27 shows XRPD patterns of Q203 free base (top trace), ND-0006E-003-04 (Type 4, middle trace) and fumaric acid (bottom trace).
- (29) FIG. 28 shows XRPD patterns of Q203 free base (top trace), ND-0006E-006-14 (Type 5, middle trace) and urea (bottom trace).
- (30) FIG. 29 shows XRPD patterns of Q203 free base (top trace), ND-0006E-006-15 (Type 6, bottom trace) and benzenesulfonic acid.
- (31) FIG. 30 shows XRPD patterns of Q203 free base (top trace), ND-0006E-003-16 (Type 7,

middle trace) and pTSA (bottom trace).

(32) FIG. **31** shows XRPD patterns of Q203 free base (top trace), ND-0006E-006-16 (Type 8, middle trace) and pTSA (bottom trace).

(33) FIG. **32** shows XRPD patterns of Q203 free base (top trace), ND-0006E-003-22 (Type 9, middle trace) and EDSA (bottom trace).

(34) FIG. **33** shows XRPD patterns of Q203 free base (top trace), ND-0006E-003-24 (Type 10, middle trace) and NDSA (bottom trace).

(35) FIG. **34** shows XRPD patterns of Q203 free base (top trace), ND-0006E-006-24 (Type 10, middle trace) and NDSA (bottom trace).

(36) FIG. **35** shows XRPD patterns of Q203 free base (top trace), ND-0006E-005-15 (Type 12, middle trace) and BSA (bottom trace).

(37) FIG. **36** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-01 (Type 13, middle trace) and 2-furoic acid (bottom trace).

(38) FIG. **37** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-03 (Type 14, middle trace) and citric acid (bottom trace).

(39) FIG. **38** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-04 (Type 14, middle trace) and fumaric acid (bottom trace).

(40) FIG. **39** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-06 (Type 16, middle trace) and ketoglutaric acid (bottom trace).

(41) FIG. **40** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-24 (Type 17, middle trace) and NDSA (bottom trace).

(42) FIG. **41** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-20 (Type 18, middle trace) and maleic acid (bottom trace).

(43) FIG. **42** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-17 (Type 19, middle trace) and gentisic acid (bottom trace).

(44) FIG. **43** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-16 (Type 20, middle trace) and pTSA (bottom trace).

(45) FIG. **44** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-13 (Type 21, middle trace) and tartaric acid (bottom trace).

(46) FIG. **45** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-12 (Type 22, middle trace) and succinic acid (bottom trace).

(47) FIG. **46** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-12 (Type 22, bottom trace) and mandelic acid (bottom trace).

(48) FIG. **47** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-07 (Type 24, middle trace) and malic acid (bottom trace).

(49) FIG. **48** shows XRPD patterns of Q203 free base (top trace), ND-0006E-005-13/20/27 (Type 25, 2.sup.nd/3.sup.rd/bottom trace).

(50) FIG. **49** shows XRPD patterns of Q203 free base (top trace), ND-0006E-007-24 (Type 26, middle trace) and NDSA (bottom trace).

(51) FIG. **50** shows XRPD patterns of Q203 free base (top trace), ND-0006E-006-17 (Type 27, middle trace) and gentisic acid (bottom trace).

(52) FIG. **51** shows XRPD patterns of Q203 free base (top trace), ND-0006E-008-15 (Type 28, bottom trace).

(53) FIG. **52** shows XRPD patterns of Q203 free base (top trace), ND-0006E-005-06 (Type 29, middle trace) and ketoglutaric acid (bottom trace).

(54) FIG. **53** shows XRPD patterns of Q203 free base (top trace), ND-0006E-005-04 (Type 30, middle trace) and fumaric acid (bottom trace).

(55) FIG. **54** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-007-28 (Type 31, bottom trace).

(56) FIG. **55** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-007-06 (Type 32,

bottom trace).

(57) FIG. **56** shows XRPD patterns of Q203 free base (top trace), ND-0006E-007-15 (Type 33, middle trace) and benzenesulfonic acid (BSA) (bottom trace).

(58) FIG. **57** shows XRPD patterns of Q203 free base (top trace), ND-0006E-007-22 (Type 34, middle trace) and ethanedisulfonic acid (EDSA) (bottom trace).

(59) FIG. **58** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-007-18 (Type 35, bottom trace).

(60) FIG. **59** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-004-18 (Type 36, bottom trace).

(61) FIG. **60** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-008-13 (Type 37, bottom trace).

(62) FIG. **61** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-009-07 (Type 38, bottom trace).

(63) FIG. **62** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-009-06 (Type 39, bottom trace).

(64) FIG. **63** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-008-10 (Type 40, bottom trace).

(65) FIG. **64** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-008-09 (Type 41, bottom trace).

(66) FIG. **65** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-008-08 (Type 42, bottom trace).

(67) FIG. **66** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-008-02 (Type 43, bottom trace).

(68) FIG. **67** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-008-01 (Type 44, bottom trace).

(69) FIG. **68** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-010-04 (Type 45, bottom trace).

(70) FIG. **69** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-010-05 (Type 46, bottom trace).

(71) FIG. **70** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-010-13 (Type 47, bottom trace).

(72) FIG. **71** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-003-34 (Type 48, bottom trace).

(73) FIG. **72** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-005-30 (Type 49, bottom trace).

(74) FIG. **73** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-004-33 (Type 50, bottom trace).

(75) FIG. **74** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-005-33 (Type 51, bottom trace).

(76) FIG. **75** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-004-34 (Type 52, bottom trace).

(77) FIG. **76** shows XRPD patterns of Q203 free base (top trace) and ND-0006E-005-34 (Type 53, bottom trace).

(78) FIG. **77** shows XRPD patterns of Q203 free base (top trace), ND-0006E-004-23 (Type 54, purple trace)

(79) FIG. **78** shows XRPD traces of solids isolated post pH solubility experiments

(80) Furthermore, reference is made to the following examples which are given to illustrate, not to limit the present invention.

DETAILED DISCLOSURE

Examples

Example 1

Summary

(81) A polymorph screening study of Q203 using different crystallization techniques including slurry and salt formation experiments was performed. Due to the very low solubility of the solid in various solvents, cooling, evaporative and anti-solvents experiments were not performed. Eight XRPD patterns namely A, B, C, D, E, F, G and H were identified. Patterns A and C proved to be the true crystalline forms of the di-tosylate salt. The polymorph screening re-slurry experiments were carried out starting from Form A. Form A is a neat form; it maintains its crystalline form when exposed to 100% RH at RT for 6 days. Form C is also a neat form; it can be obtained by re-slurrying Form A in methanol; it maintains its crystalline form when exposed to 100% RH at RT for 6 days. Pattern G is obtained from water and its ¹H-NMR shows that the ratio of base:acid is 1:1.3. Pattern H is also maintained its crystalline XRPD pattern when exposed to 100% RH at RT for 3 days. Pattern H is obtained from Toluene and its NMR shows that the ratio of base:acid is 3:3.46. Therefore, Patterns G and H are not considered to be the true polymorphs of the di-tosylate salt. Forms A and C were stirred in six pure solvents at RT and 50° C. for 4 days. According to the XRPD scans, the residual solid from stability experiments from the majority of solvents (THF, EtOH, and IPA) was Form A, but methanol as the re-slurry medium generated Form C. According to the data, Form A is considered to be more stable than Form C.

(82) XRPD

(83) Details of XRPD method used in the tests are mentioned below: Rigaku D/MAX 2200 X-ray powder diffract meter X-ray Generator: Cu, K α , (λ =1.54056 Å) Tube Voltage: 40 kV, Tube Current: 40 mA DivSlit: 1 deg DivH.L.Slit: 10 mm SctSlit: 1 deg RecSlit: 0.15 mm Monochromator: Fixed Monochromator Scanning Scope: 3-36 deg (2-theta) Scanning Step: 5 deg/min

DSC

(84) Details of DSC method used in the tests are mentioned below: Mettler Toledo Q2000 MDSC Heat from 40° C. to 300° C. at 10° C./min

Experiments

(85) Analysis of the Starting Material

(86) XRPD, DSC and TGA scan of the starting di-tosylate salt from A are shown in FIGS. 1, 4 and 5. According to XRPD, the solid (named as Form A) is crystalline; the DSC scan of the solid shows a melting onset temperature of around 235° C.; TGA data indicates a weight loss of 0.3% from 30° C.-200° C. The purity of the starting material is 99.46%.

Solubility Experiments

Exp 1: Solubility of the Free Base

(87) The objective of measuring solubility of free base was to identify potential solvents for salt formation process development. Approximate solubility of the free base was measured by gravimetry method. Excess solids were added to various solvents including some Class III solvents (ICH guideline) at RT and 50° C. Since the solubility in these solvents is generally low, it was decided to mix water with these solvents. The suspensions were slurried for one day; the clear liquors from the top portion of the suspension were used for solubility determination by gravimetry method. The solubility data are given in Table 2-1. At 50° C., the solubility in IPA or its mixture with water is generally low (11-22 mg/ml), in acetone and its mixture with water is within 41-74 mg/ml range, in EA and its mixture with water is 55-98 mg/ml range, and in methyl acetate is 94-105 mg/ml range. The solubility in THF was 279 mg/ml at 50° C.

(88) TABLE-US-00001 TABLE 2-1 Approximate solubility* data of the free base in pure solvents at RT and 50° C. Solubility (mg/ml) Solvents RT 50° C. Ethyl Acetate (EA) 13 99 EA + 4% water 16 55 Acetone N/A 48 Acetone + 5% water 22 74 Acetone + 10% water N/A 42 Isopropyl Alcohol (IPA) N/A 20 IPA + 5% water N/A 22 IPA + 15% water N/A 11 Isopropyl Acetate (IPAC) 26 51 IPAC + 4% water N/A 35 Tetrahydrofuran (THF) 133 279 Methyl Amine (MA) 38 94 MA + 4% water 18 105 *The solubility numbers are approximate and are intended for process development

(97) TABLE-US-00004 TABLE 4-1 Results of starting material re-slurry in binary solvents (50:50, vol:vol) at RT for 5 days Water MeOH EtOH MA EA acetone MeCN THF IPA IPAC TBME MEK DCM Toluene Water MeOH B EtOH F A MA C A EA C D A Acetone F C A A A MeCN N/A C F F F F THF A A A A A A A IPA B C A N/A F A A A IPAC F A A A A A A A MTBE A A A A A A A A MEK A A A A A F A A A A DCM A A A A A A A A A A A Toluene A A A A A A A A A A A MIBK C A A A A A A A A A A A

Salt Formation Experiments

(98) In order to screen conditions other than slurry experiments, salt formation (reactive crystallization) experiments in 12 solvents were carried out. The procedure of the experiments was shown below. 150 mg free base was added in a 4 ml vial with magnetic stirrer 10 vol (1.5 ml) solvents (the list of the solvents are shown in Table 4 were added The solutions were stirred at 50° C. for 2 hours 2.2 eq mol (113 mg) of the P-toluenesulfonic acid were added into the solutions at 50° C. As soon as suspension with sufficient amount of solid (for XRPD/DSC test) were formed, samples were filter/dried and were subjected to XRPD/DSC analysis Solids with new XRPD patterns were analyzed by NMR for acid/base ratio

(99) Table 5-1 shows observations during salt formation experiments. It is noticed that the free base dissolved in the target solvents at 50° C. before counter ion addition. In all cases, the solids (potential salts) are formed almost immediately after counter ion addition. The solids at this point (immediately after salt formation) were analyzed before and after drying. Wet and dry XRPD analyses help to identify potential solvates/hydrates of the salt. In some wet solids, some deviation from pattern C was observed, however the dry solids in all solvents except in toluene generated pattern C. The major observation from the experiment is the unanimous exhibition of appearance of pattern C immediately after salt formation. It is known that pattern C eventually converts to pattern A in solvents (except in methanol).

(100) TABLE-US-00005 TABLE 5-1 observations during salt formation experiments Solvents IPA Acetone (5% (5% Process THF EtOH MeOH MA IPAC EA MEK Toluene IPA H.sub.2O) Acetone H.sub.2O) Free Base Clear Cloudy Cloudy Clear Cloudy Clear Clear Cloudy Cloudy Cloudy Clear Clear 50° C. 2 hours Add Phenomenon Solid Clear Clear Solid Clear Solid Solid Clear Clear Clear Solid Solid Acid come soon no come soon come come soon soon soon come come out then solid out then out out then then then out out quickly solid quickly solid quickly quickly solid solid solid quickly quickly come come come come come out out out out quickly quickly quickly quickly quickly XRD wet N/A C C + C + C + C H C C C C Un- Un- Un- known known known dry N/A C C C C C H C C C C Cool 1 hour N/A N/A C N/A N/A N/A N/A N/A N/A N/A N/A N/A to RT acid-base radio 5.55:3 N/A N/A N/A N/A N/A 6:3 3:3 N/A 6:3 N/A N/A (theoretical value is 6:3)

Attempt to Produce Solids with Different XRPD Patterns

(101) Various slurry and reactive crystallization experiments generated different crystalline patterns including A, B, C, D, E, F, G and H. Previous analysis (NMR) indicated that pattern H is not a true crystalline form of the di-tosylate salt, and therefore should be dropped from the list. In order to evaluate and further analysis other solids with other XRPD patterns (A-G), attempts were made to reproduce the solids (the previous experiments showing these solids had been carried out at small scales, and the majority of the solids had been consumed for XRPD/DSC analysis). Table 6-1 shows the experiment with the objective of producing solids with XRPD patterns B, C, D, E, and F. Solids with pattern A was the starting material, therefore it was already available.

(102) TABLE-US-00006 TABLE 6-1 Results of preparation of Form B, C, D, E and F Target Time Form Solvent Condition 1 day 3 days 5 days B >10 vol water RT G 50° C. RT – seed 50° C. + seed G N/A <10 vol water RT B N/A 50° C. G C MeOH RT N/A C D acetone RT A IPA EtOH MeCN RT 50° C. E MEK RT 50° C. A E A F EtOH + MeCN RT A Water + EtOH G F MA + MeCN F N/A EA + MeCN A

Analysis Various Forms

(103) FIG. 6 shows the overlay of scans of various XRPD patterns observed in this study. Patterns

A and C are true crystalline polymorphs of di-tosylate salt. Therefore the present inventors refer to this solids as Forms A and Form C. However, as other XRPD patterns have not been shown to be the true polymorph of the di-tosylate salt, the present inventors refer to them as just “pattern”. It is mentioned that patterns C and D are very similar, and one may refer to them as the same crystalline form.

(104) FIG. 7 sketches the DSC overlay associated with various XRPD patterns of the solid. The onset temperature for Form A, C (and D), and pattern E are the same and varies between 235° C. to 240° C. Pattern B shows double endothermic peaks, one with an onset temperature around 139° C. and the other one around 208° C.

(105) Analysis of Form A

(106) Form A is a true polymorph of the di-tosylate salt; NMR shows the acid: base ratio is 6:3. The crystalline is generally low, the DSC scan shows a melting onset around 235° C.; the TGA scan shows 0.3% weight loss from 30° C. to 200° C. (FIGS. 4-5). It is the predominant form obtained in slurry experiments. Conversion from Form A to other XRPD patterns was mainly obtained from methanol (to Form C) or water (to pattern B) in the slurry medium. However, as it will be shown in further sections, Form C is converted to Form A upon re-slurry in the majority of other solvents (than methanol). Conversion of Form A to pattern B solids do not imply that solid with pattern B is a more stable form than Form A, since it is highly possible that pattern B is a partial (or full) hydrolyzed species of the salt. FIG. 8 shows the DVS of Form A, the water absorption is around 0.35% at 90% RH at RT.

(107) Analysis of Form C

(108) FIG. 9 to 11 show XRPD, DSC, and TGA of a solid with Form C. Form C is a true polymorph of the di-tosylate salt as NMR showed the acid: base ratio is 2:1. The XRPD peaks of pattern C are sharper than those of Form A. Form C is the predominant form when methanol is used as the slurry medium. However, Form “C” is not a methanol solvate form of the di-tosylate salt. Many reactive crystallization experiments using different solvents generated Form C immediately after salt formation. Later experiments showed that overnight slurry of the same suspensions (reactive crystallization) transform Form C to A. FIG. 12 shows the DVS scan of Form C. The DVS scan shows that the solid only takes less than 0.3% water at 90% RH at RT.

(109) Analysis of Form G

(110) XRPD, DSC, and TGA of a solid with pattern G are shown in FIGS. 13 to 15. The solid is crystalline, it shows a DSC melting onset of 216° C. (lower than those of Form A and C), exhibited weight loss of 0.3% in the range of 30° C.-200° C. The HPLC data shows the acid to base ratio to be 3.45:1, therefore it is not considered to be the di-tosylate salt.

(111) Other Patterns

(112) The efforts made to reproduce other XRPD patterns did not generate these patterns, meaning that these solids were only observed once or twice during form screening. Among these pattern B is significant, because it may point to a hydrate form. However, it is probably a hydrated form of the hydrolyzed salt, not the di-tosylate salt (hydrate of partial salt).

Stability Experiments

(113) The objective of the work in this section is to observe the potential transformation of these forms when exposed to 100% RH as well as to identify the most stable polymorph observed in this study.

Exp 1: Exposing Forms a and C in 100% RH Experiment at RT for 6 Days

(114) Forms A and C were exposed in 100% RH environment at RT for 6 days; the solids were subsequently analyzed using DSC and XRPD.

(115) The results were shown in FIG. 16 to 21. In FIGS. 1 and 17, the XRPD of sample of Form A (previous obtained form EA and IPA/MEK) before and after 6 days exposure to 100% RH are sketched. FIGS. 19 and 20 show the DSC scan of the same samples after exposure. Both figures indicated no change to crystalline, shows Form A is stable (from polymorph point view) in high RH

environment. FIGS. 18 and 21 show the result of the same treatment for Form C. The results show that the crystal structure does not change as a result of exposure to high RH.

Exp 2: Forms A and C Re-Slurry in 6 Different Pure Solvents at RT for 5 Days (Aging Experiments)

(116) These experiments were performed to investigate potential transformation between Form A and C. The procedure of the experiments is shown below. 15 mg Form A and 15 mg Form C were add into a 1.5 ml vial with magnetic stirrer. 20 vol (600 ul) of various solvents was added into the vial The suspensions were stir at RT for 5 days Then filter and the solid was sent for XRPD (before and after drying in oven) and DSC

(117) The experimental data are shown in Table 10-1. Water, methanol, IPA, acetone, and THF were used as the slurry medium. The data from day 5 samples indicated partial hydrolysis of the ditosylate salt in water (producing pattern G solid). One more times methanol as the slurry medium produced Form C, which is consistent with previous re-slurry findings. These data indicate that Form A is more stable than Form C (except in methanol) and thus appears to be the most stable form under most conditions.

(118) TABLE-US-00007 TABLE 10-1 Results of Forms A and C re-slurry in 6 different pure solvents at RT for 5 days RT 4 days 5 days Solvents wet dry wet dry Water G G G G MeOH C C C C EtOH A A A A IPA Unknown A A A THF A A A A

Exp 3: Forms A, C, and H Re-Slurry in 6 Different Pure Solvents at RT and 50° C. For 3 Days

(119) The procedures for those experiments were shown below. 10 mg Form A and 5 mg Form C were added into a 1.5 ml vial with magnetic stirrer. 20 vol (300 ul) of various solvents was added into the vial and it was ensured that the solution was cloudy Add little Form H as seed The suspensions were stirred at RT for 3 days Then filter and the solid was sent for XRPD (after drying in oven) and DSC

(120) The results are shown in Table 11-1 except for water (partial hydrate) and methanol (From C), other treatment at both temperatures give Form A (as expected).

(121) TABLE-US-00008 TABLE 11-1 results of the stability experiments of Forms A, C, and H Form solvents RT 50° C. IPA A A THF A A acetone A A MeOH C N/A EtOH A A water H H

Exp 4: the Stability of Form C Through Seeding Experiments

(122) Salt formation screening (reactive crystallization experiments) produced Form C. To evaluate the polymorph stability of Form C, a small amount of Form A was added into the suspensions of Form C (in the below solvents) and stirred overnight at RT. The suspensions were filtered and the solids were analyzed using XRPD (wet and dry sample). The results are shown below. The data indicate that Form C converts to Form A in all solid, confirming previous results on stability of Form A.

(123) TABLE-US-00009 TABLE 12-1 results of the stability experiments RT solvents

Wet(overnight) Dry(overnight) IPA A A IPA + 5% water A A MEK A A MA A A Acetone A A Acetone + 5% water A A

Example 2

(124) Inventors wanted to perform a salt/co-crystal screen of Q203 free base, shown below. To this end, a standard salt/co-crystal screen of Q203 was carried out in order to identify salts with acceptable properties and also physicochemical characterisation of Q203 free base was conducted. The starting material used in this example 2 were Q203 free base, batch C12032302-J16001 and a ditosylate salt of Q203, batch C12032302-K16001M.

(125) ##STR00002## Chemical Formula: C.sub.29H.sub.28ClF.sub.3N.sub.4O.sub.2 Molecular Weight: 557.01 Elemental Analysis: C, 62.53; H, 5.07; Cl, 6.36; F, 10.23; N, 10.06; O, 5.74 Q203 free base chemical structure

(126) The following co-formers, listed in Table 1, have been used for the salt/co-crystal of Q203 free base:

(127) TABLE-US-00010 TABLE 1 List of co-formers used in this example 2 Co-formers 2-furoic

acid isethionic pantoic acid hydriodic acid oxalic acid (2-hydroxy- ethanesulfonic) acid ascorbic acid ketoglutaric phosphoric hydrochloric acid nitric acid (oxoglutaric) acid citric acid lactobionic p- glycolic naphthalene-2- acid toluenesulfonic (hydroxyacetic) sulfonic acid acid acid estolic maleic acid pyroglutamic glucoheptonic mono-methyl (laurylsulfonic) (L) acid acid ester of sulfonic acid acid ethane-1,2- malic (L) acid pyruvic (2- glucaric methanesulfonic disulfonic acid oxopropanoic) (saccharic) acid acid acid fumaric acid malonic acid saccharin gentisic (2,5-mandelic acid dihydroxybenzoic) acid gluconic acid tartaric (L) mucic acid urea sulphuric acid acid succinic acid salicylic acid — — —

2 Experimental

(128) 2.1 Solubility Estimation

(129) Aliquots of the test solvent were added to an accurately weighed sample (~25 mg) of Q203 batch C12032302-J16001 at ambient temperature. The aliquot volumes were typically 50-100 μ L. Complete dissolution of the test material was determined by visual inspection. The solubility was estimated from these experiments based on the total solvent used to provide complete dissolution. It should be noted that the actual solubility may be greater than that calculated because of the use of solvent aliquots that were too large or due to a slow rate of dissolution.

(130) If dissolution did not occur after the last aliquot of solvent was added (typically ~40 volumes of solvent), the sample was subjected to two cycles of the following temperature cycling regime on the Clarity crystallisation station: Heat from 20° C. to within 3° C. of solvent boiling point (or 100° C., whichever was lower) at 0.5° C./minute. Cool to 20° C. at 0.2° C./minute. Stirrer speed 800 rpm.

(131) From the infrared (IR) transmission data of the sample vials, dissolution and precipitation events were recorded as the point of complete transmission of IR and the onset of turbidity by IR respectively. The solubility values for Q203 were expressed as a range and rounded to the nearest whole number.

(132) 2.2 Screening Methods

(133) Experiments were carried out at a scale of ~25 mg with 1:1 stoichiometry and 2:1 stoichiometry (salt/co-crystal former: Q203 free base).

(134) 2.2.1 Slow Evaporation

(135) A stock solution of coformer in a chosen solvent (1 eq.) was added to a stock solution of Q203 free base (Batch C12032302-J16001, 1 or 2 eq.). Where stock solutions of the coformer were not made, the coformer was added as a solid/liquid. The resultant solution was evaporated in a fume hood at ambient temperature in a vial covered with perforated aluminium foil. Isolated solids were dried under nitrogen then analysed by XRPD.

(136) 2.2.2 Slurry Experiments

(137) Q203 free base (Batch C12032302-J16001, 1 eq.) and coformer (1 or 2 eq.) was added to a given solvent until undissolved solids remained at the desired temperature (20 or 40° C.). The vial was sealed and the slurry was maintained at the selected temperature and agitated by magnetic stirring for 5-7 days. Solids were isolated by centrifugation and liquid decantation and dried under nitrogen prior to analysis by XRPD.

(138) 2.2.3 Sonication

(139) A selected neat or mixed solvent system was added to Q203 free base (Batch C12032302-J16001, 1 eq) and coformer (1 eq) to form a paste. The paste was sonicated at 70% intensity using a Cole-Parmer 130W ultrasonic processor using a pulsed program. All solids recovered from these experiments were dried under nitrogen prior to analysis by XRPD.

(140) 2.2.4 Liquid Assisted Grinding (LAG)

(141) Q203 free base (~50 mg, 1 eq.) was added to a stainless steel milling chamber with selected co-former (1 eq.). A milling ball and solvent (25 μ L) was added to the chamber, which was milled for 3×2 mins @25 Hz, scraping the walls of the chamber in between each run. The grinded solid was analysed by XRPD.

(142) 2.2.5 Co-Melting (Kofler Melt)

(143) Q203 free base (~50 mg, 1 eq.) was added to a HPLC vial with selected co-former (1 eq.). The vial was pre-purged with nitrogen and the temperature of the hotplate was increased until one solid melted and diffused into the other. The molten material was allowed to cool to ambient prior to subsequent analysis by XRPD.

(144) 2.2.6 Humidity Stress of Generated Salts

(145) Approximately 25 mg of generated salts of Q203 were added to vials and placed unsealed into a 75% relative humidity chamber (sealed cabinets with relative humidity conditions controlled by super-saturated salt solutions) at ambient temperature for 7 days prior to analysis by XRPD.

2.3 Experimental Techniques

(146) 2.3.1 X-ray Powder Diffraction (XRPD)

(147) XRPD analyses were performed using a Panalytical Xpert Pro diffractometer equipped with a Cu X-ray tube and a Pixcel detector system. The isothermal samples were analysed in transmission mode and held between low density polyethylene films. 2 XRPD programs were used (range 3-40° 2 θ , step size 0.013°, counting time 99 sec, ~22 min run time and range 3-40° 2 θ , step size 0.013°, counting time 46 sec, ~11 min run time). XRPD patterns were sorted and manipulated using HighScore Plus 2.2c software.

(148) 2.3.2 Differential Scanning Calorimetry (DSC)

(149) DSC analyses were carried out on a Perkin Elmer Jade Differential Scanning Calorimeter. Accurately weighed samples were placed in crimped aluminium pans. Each sample was heated under nitrogen at a rate of 10° C./minute to a maximum of 300° C. Indium metal was used as the calibration standard. Temperatures were reported at the transition onset to the nearest 0.01 degree. Note that DSC traces within this report may contain automated peak integrations which calculate ΔH of fusion. Where multiple thermal events are observed at similar temperatures, these ΔH values are prone to significant error.

(150) 2.3.3 Thermogravimetric Differential Thermal Analysis (TG/DTA)

(151) Thermogravimetric analyses were carried out on a Mettler Toledo TGA/DSC1 STARe. The calibration standards were indium and tin. Samples were placed in an aluminium sample pan, inserted into the TG furnace and accurately weighed. The heat flow signal was stabilised for one minute at 25° C., prior to heating to 300° C. in a stream of nitrogen at a rate of 10° C./minute.

(152) 2.3.4 .sup.1H/.sup.13C Nuclear Magnetic Resonance spectroscopy (NMR)

(153) NMR analysis was carried out on a Bruker 500 MHz instrument in MeOD-d.sub.4 or DMSO-d.sub.6. Instrumental parameters are listed on the relevant spectrum plots.

(154) 2.3.5 Optical Microscopy

(155) Microscopy analyses were carried out using an Olympus BX51 stereomicroscope with crossed-polarised light and a 1st order red compensator plate. Photomicrographic images were captured using a ColorView IIIu digital camera and SynchronizIR basic V5.0 imaging software with objective lens magnification of $\times 10$.

(156) 2.3.6 HPLC

(157) HPLC was used to determine equilibrium solubility in a variety of solvents. Samples in solvents that were immiscible with water were evaporated to dryness and re-dissolved in sample diluent.

(158) 2.3.7 Solubility Determination

(159) To determine the solubility suspensions of Q203 salts (p-TSA, phosphate & HCl) in the chosen pH conditions (1, 4.5, 6.8 & 7.5), experiments were set up in duplicate. Salt was weighed into vial (~25 mg) and aliquot of chosen buffer was added (1 mL). Suspension were placed on the plate and stirred at room temperature for 22-72 hours. The pH of samples were monitored during the stirring and adjusted to maintain pH within ± 0.5 pH units of the parent buffer solution. At the end of the experiment the aliquot was withdrawn at the end of the experiment and filtered through syringe PTFE filter (0.45 μ m), pH was checked and the solution was injected onto the HPLC

system without dilution. If results showed samples were too concentrated, samples were diluted and rerun. The solids recovered were analysed by XRPD and checked for form change.

(160) The preparation methods for the solutions used in this study are shown below:

(161) 2.3.8 Components for Standard Buffer Solutions Preparation (According to USP 27)

(162) 2.3.8.1 Potassium Chloride 0.2M

(163) Potassium chloride (1.5 g) was weighed into 100 mL flask and filled to the mark with water.

(164) 8.2.1.2 Monobasic Potassium Phosphate 0.2M

(165) Monobasic potassium phosphate (2.8 g) was weighed into 100 mL flask and filled to the mark with water.

(166) 2.3.8.2 Sodium Hydroxide 0.2M

(167) Standard solution of NaOH (2.0M, 10 mL) was added to 100 mL flask and filled to the mark with water.

(168) 2.3.8.3 Potassium Biphthalate 0.2M

(169) Potassium biphthalate (4.1 g) was added to 100 mL flask and filled to the mark with water.

(170) 8.2.1.4 Hydrochloric Acid 0.2M

(171) HCl Standard solution (1.0M, 20 mL) was added to 100 mL flask and filled to the mark with water.

(172) 2.3.8.4 USP Buffer pH 1

(173) 0.2M Potassium chloride solution (50 mL)+0.2M HCl solution (85 mL) was added to 200 mL flask and filled to the mark with water.

(174) 2.3.8.5 USP Buffer pH 4.5

(175) 0.2M Potassium biphthalate solution (50 mL)+0.2M NaOH solution (6.6 mL) was added to 200 mL flask and filled to the mark with water

(176) 2.3.8.6 USP Buffer pH 6.8

(177) 0.2M monobasic Potassium phosphate solution (50 mL)+0.2M NaOH solution (22.4 mL) was added to 200 mL flask and filled to the mark with water.

(178) 2.3.8.7 USP Buffer pH 7.5

(179) 0.2M monobasic Potassium phosphate solution (50 mL)+0.2M NaOH solution (39.1 mL) was added to 200 mL flask and filled to the mark with water.

(180) Buffers were adjusted with 1.0M HCl and 2.0M NaOH to reach correct pH if necessary.

(181) 2.3.9 pKa Analysis

(182) The sample pKas were determined using the spectrometric (UV-metric) technique. The sample was titrated in a UV-metric triple titration from pH 2.0-12.0 at concentrations of 31-23 μ M, under methanol-water co-solvent conditions (the methanol mixing ratio varied from 63.9 to 46.7% w/w).

3 Characterisation and Solvent Screen on Q203

(183) 3.1 pKa Determination

(184) For pKa determination, no precipitation of the sample from solution was observed and two pKas, with aqueous values of 3.70 ± 0.06 and 4.97 ± 0.01 determined from the spectroscopic data collected by Yasuda-Shedlovsky extrapolation of the individual results obtained (see Table 2).

(185) It should be noted that an additional potentiometric assay was carried out which confirmed the pKas and that there were no further pKas associated with the sample within the measurable pH range (2.0-12.0).

(186) TABLE-US-00011 TABLE 2 pKa results of Q203 free base pKa T/ $^{\circ}$ C. Ionic environment Method 3.70 ± 0.06 25.0-25.1 0.15M KCl UV-metric 4.97 ± 0.01 25.0-25.1 0.15M KCl UV-metric

3.2 Characterisation of Q203 Free Base

(187) The XRPD pattern obtained for Q203 free base batch C12032302-J16001, is shown in FIG. 22. The XRPD pattern is indicative of a highly crystalline material. Proton NMR analysis of Q203 free base showed the material conformed to the molecular structure, with a likely presence of residual solvent (potentially acetone $\sim$ 80 ppm) (data not shown).

(188) 3.3 Characterisation of Q203 Ditosylate Salt

(189) The XRPD pattern obtained for Q203 ditosylate batch C12032302-K16001M, is shown in FIG. 23 (Form A or “pattern A”). The XRPD pattern is indicative of a crystalline material, with a slight elevation of the baseline and broadening of peaks suggesting possible amorphous content. Proton NMR analysis of Q203 ditosylate showed the material conformed to the molecular structure with a 2:1 acid/API stoichiometry (data not shown).

(190) 3.4 Estimated Solubility of Q203 Free Base

(191) The solubility of Q203 free base was estimated in 10 solvent systems using the aliquot addition method. Temperature cycling experiments were also performed using the Clarity crystallisation station in order to assess the solubility of the compound with heating. The solubility data are detailed in Table 3. Observations recorded for each experiment are described in Table 3. Q203 free base was found in 4 solvents at room temperature at ~25 mg/mL and in further 4 other solvents upon heating. It was shown no dissolution upon temperature in acetone and MTBE.

(192) TABLE-US-00012 TABLE 3 Solubility estimates of Q203 free base at 20° C. Solubility range T.sub.disso T.sub.cryst Solvents Acronym (mg/mL) (° C.) (° C.) acetone — <25.4 *35 *not observed acetonitrile ACN <25.0 58.2-58.9 47.4-47.9 dimethyl sulfoxide DMSO <25.5 53.6-54.7 32.8-39.3 ethyl acetate EtOAc 34-36 ambient not observed IPA IPA <25.3 52.5-55.7 37.0-39.3 methanol MeOH <25.0 45.6-47.1 25.3-26.6 MEK MEK 50-56 ambient not observed methyl tert-butyl MTBE <25.6 not not ether observed observed tetrahydrofuran THF 128-255 ambient not observed dichloromethane DCM 169-253 ambient not observed *= showed partial solubility when heated however solid remained after two Clarity heat/cool cycles

3.5 Conclusions from Characterisation and Solvent Screening

Q203 Free Base XRPD analysis indicated that Q203 free base (“free base” herein also sometimes abbreviated as “FB”) batch C12032302-J16001, is indicative of a highly crystalline material. Proton NMR analysis of Q203 free base (data not shown) showed the material conformed to the molecular structure, with a likely presence of residual solvent (potentially acetone ~80 ppm). TG/DTA data showed ~0.4% of weight loss ~40° C. to 235° C. suggesting minimal moisture or solvent content, indicating Q203 free base, batch C12032302-J16001 to be an anhydrous material with some residual moisture/solvent. A second weight loss at temperatures greater than 235° C. corresponds to the initiation of decomposition of the material. A melting endotherm was observed at onset temperature 166.8° C. DSC analysis confirmed the TG/DTA results showing an endotherm event at temperature onset of ~167° C. Polarised light microscopy of Q203 free base showed the presence of individual and agglomerated particles suggesting a polydispersed PSD. pKA analysis showed two pKA's aqueous values of 3.70±0.06 and 4.97±0.01.

4 Salt/Co-Crystal Screening

(193) An extended salt/cocrystal screen has been performed on Q203 using 37 co-formers, the objective of which was to find alternative salts with more desirable properties (e.g. reduced hygroscopicity, chemical stability, dissolution rate, crystallinity, physical stability, etc.).

(194) The approach was to generate solids under a wide and diverse range of nucleation conditions, designed to mimic the process conditions and solvents used during development and formulation. A tailored, manual/semi-automated investigation conducted by experienced specialists is widely regarded as the preferred approach and has been found to perform as well as, or better, than high throughput screening while using fewer experiments.^{sup.1} A. J. Alvarez, A. Singh, A. S. Myerson, “Polymorph Screening: Comparing a Semi-Automated Approach with a High Throughput Method”, *Crystal Growth and Design*, 2009, 9, 4181-4188

(195) All solids from the crystallisation experiments were analysed by XRPD and the resulting patterns compared to that exhibited by the starting material. Novel XRPD patterns were assigned an alphabetical descriptor in order of discovery (Type 2, Type 3 etc). Where sufficient material was available, further analysis (e.g. NMR or TGA) was conducted on solids with novel XRPD patterns to allow tentative assignment of the novel pattern as a polymorph, solvate, hydrate, degradant or

mixture thereof. A summary of all the experiments performed are listed in Appendix 1, table 25, further below.

(196) 4.1 Solvent Based Screening Techniques

(197) Solvent based experiments were performed on approximately 25-40 mg scale in glass vials or in thin walled glass capillaries. The methods employed are described in detail in section 2.2.

Evaporation, slow cooling, crash cooling, crash precipitation and prolonged slurry (at ambient and elevated temperatures) mimic conditions that are likely to be encountered during process development and manufacturing. Varying the nucleation conditions in this way maximises the chance of finding new forms and also the frequency of occurrence of these forms under typical processing conditions.

(198) 4.1.1 Slow Evaporation

(199) Slow evaporation experiments were carried out as described in section 2.2.1 and the results are shown in the Table 4. XRPD analysis of resultant solids showed 14 novel Types (pure or in mixtures) from coformers including fumaric acid (Type 4), urea (Type 5), BSA (Type 6), pTSA (Types 7 and 8), EDSA (Type 9), NDSA (Types 10 and ii), saccharin (Type 25+3), gentisic (Type 27) and salicylic acid (Type 48). Types 2 and 3 were also observed both pure and in mixtures from multiple coformers. These Types are discussed further in section 5.

(200) TABLE-US-00013 TABLE 4 Screening results from slow evaporation experiments involving Q203 free base Sample No. (ND- Acid:API Salt former 0006E-) Solvent Antisolvent (molar eq.) Result XRPD 2-furoic 003-01 THF methanol 1:1 solution/solid Type 2 2-furoic 006-01 THF methanol 2:1 solid Type 2 + co-former ascorbic 006-02 THF methanol 2:1 solid Type 3 + co-former ascorbic 003-02 THF methanol 1:1 solution/solid Type 3 + co-former benzenesulfonic 003-15 THF acetone 1:1 solid Type 3 + Type 6 benzenesulfonic 006-15 THF acetone 2:1 solid Type 6 citric 006-03 THF methanol 2:1 gel amorphous citric 003-03 THF methanol 1:1 solution/solid Type 3 ethane-1,2- 003-22 THF acetone 1:1 solid Type 9 disulfonic ethane-1,2- 006-22 THF acetone 2:1 gel v. disulfonic disordered fumaric 003-04 THF methanol 1:1 solution/solid Type 4 fumaric 006-04 THF methanol 2:1 solid Type 4 galactaric 003-29 THF none 1:1 solution/solid Type 3 + (mucic) co-former galactaric 006-29 THF none 2:1 solid Type 3 + (mucic) co-former gentisic 006-17 THF methanol 2:1 solid Type 27 gentisic 003-17 THF methanol 1:1 solution/solid Type 3 gluconic (D) 006-25 THF methanol 2:1 solid Type 2 + Type 3 gluconic (D) 003-25 THF methanol 1:1 solution/solid Type 3 gluconic (D) 003-32 THF water 1:1 solution/solid Type 3 glycolic 006-05 THF methanol 2:1 solid Type 2 (hydroxyacetic) glycolic 003-05 THF methanol 1:1 solution/solid Type 2 + (hydroxyacetic) Type 3 HCl 003-18 THF water 1:1 solid Type 3 HCl 006-18 THF water 2:1 solid Type 3 HI 003-31 THF water 1:1 solution/solid Type 3 (disordered) isethionic 003-35 THF methanol 1:1 solution/solid Type 3 ketoglutaric 006-06 THF methanol 2:1 solid disordered (oxoglutaric) ketoglutaric 003-06 THF methanol 1:1 solution/solid Type 3 (oxoglutaric) lactobionic 003-19 THF water 1:1 solution/solid Type 3 lactobionic 006-19 THF water 2:1 solid Type 3 maleic 006-20 THF methanol 2:1 solid amorphous maleic 003-20 THF methanol 1:1 solution/solid Type 2 + Type 3 malic (L) 006-07 THF methanol 2:1 solid Type 2 + Type 3 malic (L) 003-07 THF methanol 1:1 solution/solid Type 3 malonic 006-08 THF methanol 2:1 solid Type 2 + Type 3 malonic 003-08 THF methanol 1:1 solution/solid Type 3 mandelic (DL) 003-09 THF methanol 1:1 solution/solid Type 2 + Type 3 mandelic (DL) 006-09 THF methanol 2:1 solid Type 2 + Type 3 methanesulfonic 006-23 THF acetone 2:1 gel amorphous methanesulfonic 003-23 THF acetone 1:1 solution/solid Type 3 naphtalene-1,5- 003-24 THF acetone 1:1 solid Type 10 disulfonic naphtalene-1,5- 006-24 THF acetone 2:1 solid Type 11 disulfonic nitric 003-30 THF water 1:1 solid Type 3 oxalic 006-10 THE methanol 2:1 solid disordered oxalic 003-10 THE methanol 1:1 solution/solid Type 2 + Type 3 pamoic 003-33 THF DMSO 1:1 solution/gel n/a phosphoric 006-26 THF methanol 2:1 solid Type 2 + Type 3 phosphoric 003-26 THF methanol 1:1 solution/solid Type 3 p- 003-16 THF acetone 1:1 solid Type 7 toluenesulfonic p- 006-16

THF acetone 2:1 solid Type 8 toluenesulfonic pyroglutamic 006-11 THF methanol 2:1 solid Type 2 + (L) co-former pyroglutamic 003-11 THF methanol 1:1 solution/solid Type 3 + (L) co-former pyruvic (2- 003-21 THF methanol 1:1 solution/solid Type 3 oxopropanoic) pyruvic (2- 006-21 THF methanol 2:1 solid Type 3 oxopropanoic) saccharin 006-27 THF acetone 2:1 gel n/a saccharin 003-27 THF acetone 1:1 solution/solid Type 25 + Type 3 salicylic 003-34 THF acetone 1:1 solution/solid Type 48 succinic 006-12 THF methanol 2:1 solid Type 2 + Type 3 succinic 003-12 THF methanol 1:1 solution/solid Type 3 sulphuric 006-28 THF water 2:1 solid — sulphuric 003-28 THF water 1:1 solution/solid Type 3 tartaric (L) 006-13 THF methanol 2:1 solid Type 2 + additional peak tartaric (L) 003-13 THF methanol 1:1 solution/solid Type 3 urea 003-14 THF methanol 1:1 solution/solid Type 3 + co-former urea 006-14 THF methanol 2:1 solid Type 5

4.1.2 Rt Slurry

(201) RT (ambient) temperature slurring experiments were conducted as described in section 2.2.2 using solid generated from slow evaporation experiments involving Q203 free base and coformer (1:1 eq.). Results are shown in Table 5. Novel patterns were observed from XRPD analysis of solids involving urea (Type 5), EDSA (Type 9), 2-furoic (Type 13), citric (Type 14), fumaric (Type 15), ketoglutaric (Type 16), NDSA (Type 17), maleic (Type 8), gentisic (Type 9), pTSA (Type 20), tartaric (Type 21), succinic (Type 22), mandelic (Type 23), malic (Type 24), HCl (Type 36), pamoic (Type 50), salicylic (Type 52) and MSA (Type 54). Type3 was also observed both pure and in mixtures from multiple coformers. These Types are discussed further in section 5.

(202) TABLE-US-00014 TABLE 5 Screening results from RT slurry experiments Sample No RT slurry Co-former (ND-0006E-) Solvent (20° C.) 2-furoic 004-01 MTBE Type 13 ascorbic 004-02 MTBE Type 3+ benzenesulfonic 004-15 MTBE Type 3 + Type 6 citric 004-03 MTBE Type 14 ethane-1,2-disulfonic 004-22 MTBE Type 9 fumaric 004-04 MTBE Type 15 galactaric (mucic) 004-29 MTBE Type 3 + co-former gentisic 004-17 MTBE Type 19 gluconic (D) 004-25 MTBE Type 3 gluconic (D) 004-32 MTBE Type 3 + additional peaks glycolic 004-05 MTBE Type 3+ (hydroxyacetic) HCl 004-18 MTBE Type 36 HI 004-31 MTBE Type 3 (disordered) ketoglutaric 004-06 MTBE Type 16 lactobionic 004-19 MTBE Type 3 maleic 004-20 MTBE Type 18 malic (L) 004-07 MTBE Type 24 malonic 004-08 MTBE Type 3 mandelic (DL) 004-09 MTBE Type 23 methanesulfonic 004-23 MTBE Type 54 naphthalene-1,5- 004-24 MTBE Type 17 nitric 004-30 MTBE Type 3 oxalic 004-10 MTBE Type 3 pamoic 004-33 MTBE Type 50 phosphoric 004-26 MTBE Type 3 p-toluenesulfonic 004-16 MTBE Type 20 pyroglutamic (L) 004-11 MTBE Type 3+ pyruvic (2- 004-21 MTBE Type 3 saccharin 004-27 MTBE Type 3 salicylic 004-34 MTBE Type 52 succinic 004-12 MTBE Type 22 sulphuric 004-28 MTBE Type 3 (disordered) tartaric (L) 004-13 MTBE Type 21 urea 004-14 MTBE Type 5

4.1.3 HT Slurry (40° C.)

(203) High temperature slurring experiments were conducted as described in section 2.2.2 using solid generated from slow evaporation experiments involving Q203 free base and coformer (both 1:1 and 2:1 eq.). Results are shown in Table 6. Novel patterns were observed from XRPD analysis of solids involving urea (Type 5), 2-furoic (Type 13), fumaric (Type 15), tartaric (Type 21), mandelic (Type 23), NDSA (Type 26), sulphuric (Type 31), BSA (Type 33), EDSA (Type 34) and HCl (Type 35). Types 32, 40 and 41 solid (pure and in mixtures) was observed from multiple coformers. These Types are discussed in more detail in section 5.

(204) TABLE-US-00015 TABLE 6 Screening results from slow evaporation experiments involving Q203 free base Sample No. (ND- Acid:API Co-former 0006E-) Solvent Antisolvent (molar eq.) Result XRPD 2-furoic 007-01 THF methanol 2:1 solid Type 13 ascorbic 007-02 IPA none 2:1 solid co-acid + additional peaks ascorbic 010-01 IPA water 1:1 solid Type 41 benzenesulfonic 007-15 IPA none 2:1 solid Type 33 citric 007-03 IPA none 2:1 solution n/a ethane-1,2- 007-22 IPA none 2:1 solid Type 34 disulfonic fumaric 007-04 IPA none 2:1 solid Type 15 galactaric 007-29 IPA none 2:1 solid Type 32 + (mucic) co-acid galactaric 010-14 IPA water 1:1 solid Type 41

+ (mucic) T28 gentisic 007-17 IPA none 2:1 solid Type 32 gluconic (D) 007-25 IPA none 2:1 solid Type 32 gluconic (D) 010-11 acetone none 1:1 solid Type 40 glycolic 007-05 IPA none 2:1 solid Type 32 (hydroxyacetic) glycolic 010-02 acetone none 1:1 solid Type 40 (hydroxyacetic) HCl 007-18 IPA none 2:1 solid Type 35 ketoglutaric 007-06 IPA none 2:1 solid Type 32 (oxoglutaric) lactobionic 007-19 IPA none 2:1 solid Type 32 + additional peaks lactobionic 010-07 IPA water 1:1 solid Type 41 maleic 007-20 IPA none 2:1 solid Type 32 malic (L) 007-07 IPA none 2:1 solid Type 32 malonic 007-08 IPA none 2:1 solid Type 32 malonic 010-03 acetone none 1:1 solid Type 41 + T28 mandelic (DL) 007-09 IPA none 2:1 solid Type 23 + additional peaks methanesulfonic 007-23 IPA none 2:1 solution n/a methanesulfonic 010-09 IPA water 1:1 solid Type 40 methanesulfonic 010-10 acetone none 1:1 solid Type 40 naphthalene- 007-24 IPA none 2:1 solid Type 26 1,5-disulfonic oxalic 007-10 IPA none 2:1 solid disordered (Type 2?) oxalic 010-04 acetone none 1:1 solid Type 41 oxalic 010-05 methanol none 1:1 solid Type 41 + T28 phosphoric 007-26 IPA none 2:1 solid Type 32 phosphoric 010-12 acetone none 1:1 solid Type 40 p- 007-16 IPA none 2:1 solid Pattern A toluenesulfonic pyroglutamic 007-11 IPA none 2:1 solid Type 32 (L) pyroglutamic 010-06 acetone none 1:1 solid Type 40 (L) pyruvic (2- 007-21 IPA none 2:1 solid Type 32 oxopropanoic) pyruvic (2- 010-08 acetone none 1:1 solid Type 41 oxopropanoic) saccharin 007-27 IPA none 2:1 solution n/a saccharin 010-13 IPA water 1:1 solid Type 41 + T28 succinic 007-12 IPA none 2:1 solid Type 32 sulphuric 007-28 IPA none 2:1 solid Type 31 tartaric (L) 007-13 IPA none 2:1 solid Type 21 urea 007-14 IPA none 2:1 solid Type 5

4.1.4 Sonication

(205) A selected neat or mixed solvent system was added to sufficient Q203 free base (Batch C12032302-J16001M) to form a paste. The paste was sonicated at 70% intensity using a Cole-Parmer 130W ultrasonic processor using a pulsed program. All solids recovered from these experiments were dried under nitrogen prior to analysis by XRPD. The results of these experiments are shown in Table 7. XRPD analysis of the resultant solids showed new patterns from experiments involving pTSA (Type 7), NDSA (Type 10), mandelic (Type 23), malic (Type 24), ketoglutaric (Type 29), nitric (Type 49), pamoic (Type 51) and salicylic acid (Type 53). Type 3 and 25 solid (pure and in mixtures) was observed from multiple coformers. These Types are discussed in more detail in section 5. Each Type is further detailed in section 5.

(206) TABLE-US-00016 TABLE 7 Screening results from sonication experiments Sample N° Co-former (ND-0006E-) Solvent Sonication ketoglutaric 005-06 THF Type 29 lactobionic 005-19 THF Type 3 maleic 005-20 THF Type 25 malic (L) 005-07 THF Type 24 malonic 005-08 THF Type 25 + Type 3 mandelic (DL) 005-09 THF Type 23 methanesulfonic 005-23 THF Type 3 (disordered) naphthalene-1,5- 005-24 THF Type 10 nitric 005-30 THF Type 49 oxalic 005-10 THF Type 25 + Type 3 pamoic 005-33 THF Type 51 phosphoric 005-26 THF Type 25 + Type 3 p-toluenesulfonic 005-16 THF Type 7 pyroglutamic (L) 005-11 THF Type 25 + Type 3 pyruvic (2- 005-21 THF Type 3 (disordered) saccharin 005-27 THF Type 25 salicylic 005-34 THF Type 53 succinic 005-12 THF Type 3 + unknown sulphuric 005-28 THF Type 3 (disordered) tartaric (L) 005-13 THF Type 25 urea 005-14 THF Type 3

4.2 Solid State Screening Techniques

(207) The non-solvent based (solid state) screening methods include ball milling, sublimation, melting and compression (section 2.2). These techniques mimic conditions that are likely to be encountered in large scale processing, e.g. on hot reactor walls or during drying and tableting operations. Varying the nucleation conditions in this way maximises the chance of finding new forms and also the frequency of occurrence of these forms under typical processing conditions.

(208) 4.2.1 Co-Melting (Kofler Melt)

(209) Co-melting (Kofler melt) experiments were conducted as described in section 2.2.5 using Q203 free base and coformer (1:1 eq.). Results are detailed in Table 8. These experiments generally produced amorphous or very disordered solids. XRPD analysis of solid produced from melting

Q203 free base with gluconic acid showed a novel pattern, assigned as Type 38. A novel pattern (Type 39) was also observed from melting experiments involving multiple coformers. These two Types are discussed in more detail in section 5.

(210) TABLE-US-00017 TABLE 8 Results from co-melting experiments Sample (ND- Salt former 0006E-) Result XRPD ascorbic 009-01 solid Form A + co-former galactaric (mucic) 009-10 solid amorphous gluconic (D) 009-07 solid Type 38 glycolic 009-02 solid v. disordered (hydroxyacetic) lactobionic 009-05 solid amorphous malonic 009-03 solid Type 39 phosphoric 009-08 solid amorphous pyroglutamic (L) 009-04 solid v. disordered pyruvic (2- 009-06 solid Type 39 oxopropanoic) saccharin 009-09 solid Type 39 + saccharin

4.2.2 Liquid Assisted Grinding (LAG)

(211) Co-melting (Kofler melt) experiments were conducted as described in section 2.2.5 using Q203 free base and coformer (1:1 eq.). Results are detailed in Table 9. XRPD analysis of resultant solids showed new patterns from LAG experiments involving phosphoric (Type 37) and ascorbic acid (Type 43). Types 28, 40 and 41 were observed from multiple coformers. These Types are detailed in section 5.

(212) TABLE-US-00018 TABLE 9 Results from LAG experiments Sample (ND- Salt former 0006E-) Solvent Antisolvent Result XRPD 2-furoic 008-01 acetone none solid Type 41 + T28 ascorbic 008-02 IPA water solid Type 43 citric 008-03 acetone none solid Type 40 + additional peaks galactaric 008-15 IPA water solid Type 28 (mucic) gentisic 008-08 acetone none solid Type 42 gluconic (D) 008-12 acetone none solid Type 28 + trace T3 glycolic 008-04 acetone none solid Type 40 + additional peaks (hydroxyacetic) lactobionic 008-09 IPA water solid Type 41 malonic 008-05 acetone none solid Type 40 methanesulfonic 008-11 acetone none solid amorphous oxalic 008-06 methanol none solid amorphous phosphoric 008-13 acetone none solid Type 37 pyroglutamic 008-07 acetone none solid Type 40 + Form A + co-former (L) pyruvic (2- 008-10 acetone none solid Type 40 oxopropanoic) saccharin 008-14 IPA water solid Type 41 + T28

4.3 Conclusions from Salt/Co-Crystal Screening

(213) Approximately 20 T experiments were carried out using solvent and non-solvent based techniques. 2 crystalline XRPD patterns were observed during this study, including the as received material. This showed a high tendency of Q203 to generate new patterns from various co-formers and techniques. Polymorphism of the Q203 was observed but most of the new patterns are likely attributed to salts and/or potential co-crystals of Q203 free base.

(214) TABLE-US-00019 TABLE 10 Summary of the observed solids of Q203 (sorted out by co-former) Comments/ oType Co-former Tentative Assignments 13 2-furoic Crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests no salt formation Potential co-crystal or polymorph of Q203 free base 43 ascorbic Disordered crystalline, prepared by LAG using a mixture of IPA/water Proton NMR: no PS, ~0.8 molar eq. co-former, no residual IPA 6 benzenesulfonic Disordered crystalline, prepared by slow evaporation using THF/acetone .sup.1H NMR suggests salt formation with ~0.25 molar eq. of THF Q203 benzenesulfonate salt (1:1 or 2:1 stoichiometry) 12 benzenesulfonic Crystalline, prepared by sonication using THF .sup.1H NMR suggests salt formation with ~0.03 molar eq. of THF Q203 benzenesulfonate (1:1 stoichiometry) 33 benzenesulfonic Crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with no residual solvent Q203 benzenesulfonate salt (likely 2:1 stoichiometry) 14 citric Crystalline, prepared by RT slurring for 7 d using MTBE .sup.1H NMR suggests no salt formation Potential co-crystal or polymorph of Q203 free base 9 ethane-1,2-disulfonic Disordered crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests salt formation with ~0.07 molar eq. of MTBE Q203 edisylate (potential 1:1 stoichiometry) 34 ethane-1,2-disulfonic Disordered crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with no residual solvent Q203 edisylate salt (2:1 stoichiometry) 4 fumaric Crystalline, prepared by slow evaporation using a mixture of THF/MeOH .sup.1H NMR suggests no salt formation and ~0.5 eq residual MeOH Potential MeOH hemi-solvate of co-crystal or

polymorph of Q203 free base 15 fumaric Crystalline, prepared by RT/HT slurring for 7 d using MTBE/IPA .sup.1H NMR suggests no salt formation (~0.7 molar eq. fumaric acid) Potential co-crystal or polymorph of Q203 free base 30 fumaric Crystalline, prepared by sonication using THF .sup.1H NMR shows no peak shifting, no residual solvent and ~0.9 molar eq. of fumaric acid suggesting possible co-crystal or polymorph of Q203 free base. 28 Galactaric, gluconic Crystalline, prepared by LAG experiments using a mixture of IPA/water and acetone from galactaric and gluconic respectively .sup.1H NMR suggests no salt formation (~0.08 molar eq. IPA). Likely polymorph of the Free base 19 gentisic Crystalline (similar to T3), prepared by RT slurring for 7 d using MTBE .sup.1H NMR suggests no salt formation (~0.1 eq residual MTBE and 0.2 eq. free acid) Potential co-crystal or polymorph of Q203 free base 27 gentisic Disordered crystalline, prepared by slow evaporation using a mixture THF/methanol .sup.1H NMR shows no peak shifting, with ~0.5 molar eq. of THF, ~2 mol. eq. of co-former, suggesting possible THF hemisolvate of co-crystal or polymorph of the free base. 42 gentisic Disordered crystalline, prepared by LAG using a mixture of IPA/water 38 gluconic Crystalline, prepared by co-melt Proton NMR: no PS as expected, no solvent, no presence of co-former TG/DTA: melt at ~165° C. Likely polymorph of the free base 35 HCl Disordered crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with no residual solvent Q203 HCl salt. The stoichiometry has not been determined 36 HCl Crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with ~0.007 molar eq. of MTBE Q203 HCl salt. The stoichiometry has not been determined 55 HCl Crystalline with the presence of disordered material, prepared by evaporation from THE Proton NMR: shifting confirming salt formation. 56 HCl Crystalline with the presence of disordered material, generated as part of the pH profile analysis from Type 55 and phosphate salt at pH1 Proton NMR: shifting confirming salt formation 16 ketoglutaric Crystalline, prepared by RT slurring for 7 d using MTBE (oxoglutaric) .sup.1H NMR shows possibly PS at 2.9 ppm, 0.02 eq residual MTBE. Nature of this Type needs to be confirmed 29 ketoglutaric Disordered crystalline, prepared by sonication using THF (oxoglutaric) .sup.1H NMR shows no peak shifting, with ~0.16 molar eq. of THF and ~0.9 molar eq. of ketoglutaric acid suggesting possible co-crystal or polymorph of Q203 free base. 18 maleic Crystalline (similar to T19), prepared by RT slurring for 7 d using MTBE .sup.1H NMR suggests salt formation and ~0.2 eq residual MTBE Potential Q203 maleate (1:1 stoichiometry), possible non-stoichiometric MTBE solvate 24 malic (L) Crystalline, prepared by RT slurring using MTBE and by sonication using THE .sup.1H NMR shows no peak shifting, no residual solvent and ~1 molar eq. of malic acid suggesting possible co-crystal or polymorph of the free base. 23 mandelic (DL) Crystalline, prepared by RT and HT slurring using MTBE and IPA, and by sonication .sup.1H NMR shows no Peak shifting and ~0.8 molar eq. of mandelic acid suggesting possible co-crystal or polymorph of the free base. 54 methanesulfonic Disordered crystalline, prepared by RT slurry for 7 d using MTBE Very similar to Type 3 Proton NMR: peak shifting with ~0.18 molar eq. of MTBE suggesting likely MSA salt of Q203 with a 1:1 stoichiometry. 2 multiple co-formers Crystalline, prepared by slow evaporation using THF/MeOH .sup.1H NMR suggests no salt formation and no visible degradation Potential polymorph of Q203 free base 3 multiple co-formers Crystalline, prepared by a range of techniques and solvents 1H NMR suggests no salt formation. Potential polymorph of Q203 free base 25 multiple co-formers Crystalline, prepared by multiple solvents and techniques obtained as pure or as a mixture with Type 3. This suggests polymorph of the free base. But .sup.1H NMR analysis of material from maleic suggests salt formation with residual solvent (~0.5 molar eq THF). Likely a THF hemi-solvate of the salt of Q203. This may be explained by possible salt formation in-situ during NMR preparation sample. 32 multiple co-formers Crystalline, prepared by HT slurring for 7 d using IPA and stoichiometry .sup.1H NMR shows no peak shifting, no residual solvent and ~0.2 molar eq. of ketoglutaric acid. As obtained from multiple co-formers, likely a polymorph of Q203 free base. 39 multiple co-formers Crystalline, prepared from various co-formers (malonic, pyruvic and saccharin) by co-melting

technique suggesting polymorph of the free base Proton NMR of saccharin and pyruvic solids showed salt formation suggesting salt formation had occurred during NMR sample preparation. 40 multiple co-formers Crystalline, prepared from various co-formers (gluconic, malonic, pyruvic. . .) and various techniques co-melting technique suggesting polymorph of the free base 41 multiple co-formers Crystalline, prepared from various co-formers (ascorbic, lactobionic, galactaric) and techniques (HT slurry, LAG) Likely polymorph of the free base 1 n/a material, crystalline, free based .sup.1H NMR analysis showed no visible degradation and trace of residual solvent (may be acetone around ~80 ppm). 10 naphthalene-1,5- Crystalline, prepared by a range of techniques and solvents disulfonic .sup.1H NMR suggests salt formation and ~0.5 molar eq. residual THF Potential THF hemisolvate of Q203 naphthalenedisulfonate (2:1 stoichiometry) 11 naphthalene-1,5- Crystalline (similar to T10), prepared by slow evaporation using THF disulfonic .sup.1H NMR suggests salt formation and ~0.7 molar eq. residual THF Potential THF solvate of Q203 naphthalenedisulfonate Stoichiometry not determined due to the presence of free acid 17 naphthalene-1,5- Crystalline (similar to T11), prepared by RT slurrying for 7 d using MTBE disulfonic .sup.1H NMR suggests salt formation with ~0.08 molar eq. of MTBE Potential Q203 NDSA salt (2:1 stoichiometry) 26 naphthalene-1,5- Crystalline, prepared by slow evap, followed by HT slurrying for 7 d using disulfonic IPA .sup.1H NMR suggests salt formation (~0.04 molar eq. IPA). The stoichiometry needs to be confirmed 49 nitric Disordered crystalline, prepared by sonication using THF Very similar to Type 3 + additional peaks Proton NMR: peak shifting with ~0.25 molar eq. of THF suggesting likely Type 49 to be a nitrate salt of Q203. 45 oxalic Crystalline, prepared by HT slurry using acetone Proton NMR: peak shifting with ~0.06 molar eq. of acetone. Stoichiometry not determined. Oxalate salt of Q203. 46 oxalic Crystalline, prepared by HT slurry using methanol Proton NMR: no peak shifting. Stoichiometry not determined. Likely polymorph of the free base. 50 pamoic Crystalline, prepared by RT slurry using MTBE Similarities with Type 3 Proton NMR: no peak shifting with ~0.07 molar eq. of MTBE and ~0.65 molar eq. of co-former Possibly polymorph of Q203 free base or co-crystal. 51 pamoic Crystalline, prepared by sonication using THF Proton NMR: no PS, ~0.9 molar eq. co-former, ~0.02 molar eq. of THF Potential polymorph or co-crystal of Q203. 37 phosphoric Crystalline, prepared by HT slurry and by LAG using acetone Proton NMR showed peak shifting with ~0.09 molar eq. of acetone Suggesting Q203 phosphate salt. The stoichiometry has not been determined Pat A p-toluenesulfonic Crystalline, prepared by slow evap. then HT slurrying using IPA, XRPD pattern similar to the ditosylate salt (Pattern A) .sup.1H NMR confirmed salt formation with a 2:1 stoichiometry 7 p-toluenesulfonic Crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests salt formation with no residual solvents. Additional peaks were observed. Q203 tosylate (potential 1:1 stoichiometry) 8 p-toluenesulfonic Crystalline, prepared by slow evaporation using THF/acetone .sup.1H NMR suggests salt formation with ~0.15 molar eq. of THF Q203 tosylate (potential 2:1 stoichiometry) 20 p-toluenesulfonic Crystalline, prepared by RT slurrying for 7 d using MTBE .sup.1H NMR suggests salt formation (~0.01 molar eq. MTBE) Q203 tosylate (1:1 stoichiometry) 44 saccharin Disordered material, very similar to Type 41, prepared by LAG using a mixture of IPA/water. Proton NMR: peak shifting, ~1 molar eq. of saccharin and ~0.5 molar eq. of IPA suggesting IPA hemisolvate of saccharin salt 47 saccharin Crystalline, prepared by HT slurry using a mixture of IPA/water Proton NMR: peak shifting with trace of IPA, ~0.7 mol. eq. of co-former suggesting salt formation of Q203 saccharin salt. 48 salicylic Crystalline, prepared by HT slurry using a mixture of THF/acetone Proton NMR: no peak shifting, ~0.9 molar eq. co-former, ~0.1 molar eq. of THF and no residual acetone suggesting potential co-crystal or polymorph of the free base. 52 salicylic Crystalline, prepared by RT slurry using MTBE Similarities with Type 3 Proton NMR: no PS, ~0.4 molar eq. co-former, ~0.4 molar eq. of MTBE Likely polymorph of Q203. 53 salicylic Crystalline, prepared by sonication using THF Proton NMR: no PS, ~1.0 molar eq. co-former, ~0.04 molar eq. of THF TG/DTA: melt at ~132° C. Potential co-crystal of Q203. 22 succinic Crystalline, prepared by RT slurrying for 7 d using MTBE 1H NMR suggests salt formation (~0.01 molar eq. MTBE)

Potential Q203 tosylate (1:1 stoichiometry) 31 sulphuric Crystalline, prepared by HT slurring for 7 d using IPA. ¹H NMR suggests salt formation with no residual solvent and the presence of additional peaks (possibly degradation). The stoichiometry has not been determined 21 tartaric (L) Crystalline, prepared by RT/HT slurring for 7 d using MTBE/IPA ¹H NMR suggests no salt formation (~0.6 eq tartaric acid + unknown peak at 8.1 ppm) Potential co-crystal or polymorph of Q203 free base 5 urea Crystalline, prepared by a range of techniques and solvents. ¹H NMR suggests no salt formation, no residual solvent Potential co-crystal or polymorph of Q203 free base 5 Preparation and Characterisation of Novel Types

(215) 5.1 Pattern a (p-Toluenesulfonic)

(216) Pattern A material (ditosylate salt “form A”) was isolated from slow evaporation then HT slurry experiment using pTSA and Q203 free base (1:1 acid/API) in IPA. XRPD analysis of Pattern A (ND-0006E-003-16) showed the material was crystalline (see also figures 24+30).

(217) 5.2 Type 2

(218) Type 2 material was generated from slow evaporation experiments involving various acids and Q203 free base (both 1:1 and 2:1 acid/API) in THF/MeOH. Experimental conditions leading to pure Type 2 and Type 2 mixtures (with co-former or Type 3) are highlighted in Table 11. XRPD analysis of Type 2 solid (ND-0006E-003-01) showed the material was crystalline with some disorder present (FIG. 25). ¹H NMR analysis of Type 2 material showed no salt formation (data not shown). As this Type was obtained from different co-formers, it is likely to be a polymorph of Q203 free base.

(219) TABLE-US-00020 TABLE 11 Experimental conditions leading to Type 2 solid (pure and mixtures) Sample No. Acid:API Salt former (ND-0006E-) (molar ratio) Result XRPD 2-furoic 003-01 1:1 solution/solid Type 2 glycolic 006-05 2:1 solid Type 2 (hydroxyacetic) 2-furoic 006-01 2:1 solid Type 2 + co-former pyroglutamic 006-11 2:1 solid Type 2 + (L) co-former tartaric (L) 006-13 2:1 solid Type 2 + additional peak glycolic 003-05 1:1 solution/solid Type 2 + 3 (hydroxyacetic) mandelic (DL) 003-09 1:1 solution/solid Type 2 + 3 oxalic 003-10 1:1 solution/solid Type 2 + 3 maleic 003-20 1:1. solution/solid Type 2 + 3 malic (L) 006-07 2:1 solid Type 2 + 3 malonic 006-08 2:1 solid Type 2 + 3 mandelic (DL) 006-09 2:1 solid Type 2 + 3 succinic 006-12 2:1 solid Type 2 + 3 gluconic (D) 006-25 2:1 solid Type 2 + 3 phosphoric 006-26 2:1 solid Type 2 + 3

5.3 Type 3

(220) Type 3 material was generated from a range of experiments involving selected acids and Q203 free base (both 1:1 and 2:1 acid/API) in neat and mixed solvent systems. Experimental conditions leading to pure Type 3 and Type 3 mixtures (with co-former, Type 2, Type 6 or unknown) are highlighted in Table 12. XRPD analysis of Type 3 solid (ND-0006E-003-25) showed the material was crystalline (FIG. 26). ¹H NMR analysis of Type 3 material showed no salt formation (data not shown). As this Type was obtained from different co-formers and solvent systems, it is likely to be a polymorph of Q203 free base.

(221) TABLE-US-00021 TABLE 12 Experimental conditions leading to Type 3 solid (pure and mixtures) Sample Acid:API No. (ND- Screen (molar Salt former 0006E-) Solvent Antisolvent method ratio) Result XRPD succinic 003-12 THF methanol slow evap 1:1 solution/ Type 3 solid tartaric (L) 003-13 THF methanol slow evap 1:1 solution/ Type 3 solid gluconic (D) 003-25 THF methanol slow evap 1:1 solution/ Type 3 solid phosphoric 003-26 THF methanol slow evap 1:1 solution/ Type 3 solid sulphuric 003-28 THF water slow evap 1:1 solution/ Type 3 solid oxalic 004-10 MTBE none slurry 1:1 solid Type 3 (20° C.) gluconic (D) 004-25 MTBE none slurry 1:1 solid Type 3 (20° C.) phosphoric 004-26 MTBE none slurry 1:1 solid Type 3 (20° C.) saccharin 004-27 MTBE none slurry 1:1 solid Type 3 (20° C.) urea 005-14 THF none sonication 1:1 solid Type 3 gluconic (D) 005-25 THF none sonication 1:1 solid Type 3 citric 003-03 THF methanol slow evap 1:1 solution/ Type 3 solid ketoglutaric 003-06 THF methanol slow evap 1:1 solution/ Type 3 (oxoglutaric) solid malic (L) 003-07 THF methanol slow evap 1:1 solution/ Type 3 solid malonic 003-08 THF methanol slow evap 1:1 solution/ Type 3 solid gentisic 003-17 THF

methanol slow evap 1:1 solution/ Type 3 solid HCl 003-18 THF water slow evap 1:1 solid Type 3 lactobionic 003-19 THF water slow evap 1:1 solution/ Type 3 solid pyruvic (2- 003-21 THF methanol slow evap 1:1 solution/ Type 3 oxopropanoic) solid methanesulfonic 003-23 THF acetone slow evap 1:1 solution/ Type 3 solid malonic 004-08 MTBE none slurry 1:1 solid Type 3 (20° C.) lactobionic 004-19 MTBE none slurry 1:1 Type 3 (20° C.) pyruvic (2- 004-21 MTBE none slurry 1:1 solid Type 3 oxopropanoic) (20° C.) glycolic 005-05 THF none sonication 1:1 solid Type 3 (hydroxyacetic) lactobionic 005-19 THF none sonication 1:1 solid Type 3 HCl 006-18 THF water slow evap 2:1 solid Type 3 lactobionic 006-19 THF water slow evap 2:1 solid Type 3 pyruvic (2- 006-21 THF methanol slow evap 2:1 solid Type 3 oxopropanoic) sulphuric 004-28 MTBE none slurry 1:1 Type 3 (20° C.) (disordered) pyruvic (2- 005-21 THF none sonication 1:1 solid Type 3 oxopropanoic) (disordered) methanesulfonic 005-23 THF none sonication 1:1 solid Type 3 (disordered) sulphuric 005-28 THF none sonication 1:1 solid Type 3 (disordered) ascorbic 004-02 MTBE none slurry 1:1 solid Type 3 + (20° C.) co-former glycolic 004-05 MTBE none slurry 1:1 solid Type 3 + (hydroxyacetic) (20° C.) co-former pyroglutamic 004-11 MTBE none slurry 1:1 solid Type 3 + (L) (20° C.) co-former ascorbic 006-02 THF methanol slow evap 2:1 solid Type 3 + co-former methanesulfonic 004-23 MTBE none slurry 1:1 solid Type 3 + (20° C.) additional peaks ascorbic 003-02 THF methanol slow evap 1:1 solution/ Type 3 + solid co-former pyroglutamic 003-11 THF methanol slow evap 1:1 solution/ Type 3 + (L) solid co-former urea 003-14 THF methanol slow evap 1:1 solution/ Type 3 + solid co-former galactaric 003-29 THF none slow evap 1:1 solution/ Type 3 + (mucic) solid co-former galactaric 004-29 MTBE none slurry 1:1 solid Type 3 + (mucic) (20° C.) co-former ascorbic 005-02 THF none sonication 1:1 solid Type 3 + co-former galactaric 005-29 THF none sonication 1:1 solid Type 3 + (mucic) co-former galactaric 006-29 THF none slow evap 2:1 solid Type 3 + (mucic) co-former benzenesulfonic 003-15 THF acetone slow evap 1:1 solid Type 3 + Type 6 benzenesulfonic 004-15 MTBE none slurry 1:1 solid Type 3 + (20° C.) Type 6 succinic 005-12 THF none sonication 1:1 solid Type 3 + unknown

5.4 Type 4 (Fumaric)

(222) Type 4 material was generated from slow evaporation experiments involving fumaric acid and Q203 free base (both 1:1 and 2:1 acid/API) in THF/MeOH. XRPD analysis of Type 4 solid (ND-0006E-003-04) showed the material was crystalline (FIG. 27) and .sup.1H NMR analysis in acetone-d.sub.6 showed no salt formation with residual MeOH (~0.5 molar eq.) (data not shown). This suggests that Type 4 is a potentially a co-crystal or polymorph of Q203 free base, possibly hemisolvated with MeOH.

(223) 5.5 Type 5 (Urea)

(224) Type 5 material was generated from experiments involving urea and Q203 free base (both 1:1 and 2:1 acid/API) in a mixture of THF/MeOH, MTBE and IPA. Experimental conditions leading to Type 5 solid are highlighted in Table 13. XRPD analysis of Type 5 solid (ND-0006E-006-14) showed the material was crystalline (FIG. 28). .sup.1H NMR analysis of Type 5 material showed no peak shifting suggesting a salt has not been formed (data not shown) with no residual solvent. A DSC was carried out to determine if a Q203:urea co-crystal had been formed however the thermogram corresponded to a melt of the free base and urea. Further investigation would be required to determine whether Type 5 is a co-crystal.

(225) TABLE-US-00022 TABLE 13 Experimental conditions leading to Type 5 solid Acid:API Sample No. (molar (ND-0006E-) Solvent Antisolvent Screen method ratio) Result 004-14 MTBE none slurry (20° C.) 1:1 solid 006-14 THF methanol slow evap 2:1 solid 007-14 IPA none slow evap/slurry 2:1 solid (40° C.)

5.6 Type 6 (Benzenesulfonic)

(226) Type 6 material was isolated from a slow evaporation experiment using benzenesulfonic acid and Q203 free base (2:1 acid/API) in a mixture of THF/acetone. Type 6 was also observed as a mixture with Type 3. XRPD analysis of Type 6 solid (ND-0006E-006-15) showed the material was

disordered crystalline (FIG. 29) and .sup.1H NMR analysis (data not shown) showed peak shifting suggesting salt formation, possibly with a 1:1 or 2:1 stoichiometry (to be confirmed) with ~0.25 molar eq. of THF.

(227) 5.7 Type 7 (p-Toluenesulfonic)

(228) Type 7 material was isolated from both a slow evaporation experiment using pTSA and Q203 free base (1:1 acid/API) in a mixture of THF/acetone and a sonication experiment using THF. XRPD analysis of Type 7 solid (ND-0006E-003-16) showed the material was crystalline (FIG. 30) and .sup.1H NMR analysis showed peak shifting suggesting salt formation with no residual solvent, possibly with a 1:1 (acid:API) stoichiometry (data not shown). It should be noted the presence of additional peaks.

(229) 5.8 Type 8 (p-Toluenesulfonic)

(230) Type 8 solid was generated from a slow evaporation experiment using pTSA and Q203 free base (2:1 acid/API) in a mixture of THF/acetone. XRPD analysis of Type 8 solid (ND-0006E-006-16) showed the material was crystalline (FIG. 31) and .sup.1H NMR analysis showed peak shifting with ~0.15 molar eq. of THF suggesting salt formation, possibly with a 2:1 stoichiometry (data not shown).

(231) 5.9 Type 9 (Ethane-1,2-Disulfonic Acid)

(232) Type 9 material was generated from experiments involving EDSA and Q203 free base (1:1 acid/API). Experimental conditions leading to Type 9 solid are highlighted in Table 14. XRPD analysis of Type 9 solid (ND-0006E-006-14) showed the material was crystalline (FIG. 32). .sup.1H NMR analysis showed peak shifting with ~0.07 molar eq. of MTBE suggesting salt formation, possibly with a 1:1 stoichiometry (data not shown).

(233) TABLE-US-00023 TABLE 14 Experimental conditions leading to Type 9 solid Sample No. (ND-0006E-) Solvent Antisolvent Screen method Result 003-22 THF acetone slow evap solid 004-22 MTBE none slurry (20° C.) solid 005-22 THF none sonication solid

5.10 Type 10 (1,5-Napthalenedisulfonic Acid)

(234) Type 10 material was isolated from both a slow evaporation experiment using NDSA and Q203 free base (1:1 acid/API) in a mixture of THF/acetone and a sonication experiment using THF. XRPD analysis of Type 10 solid (ND-0006E-003-24) showed the material was crystalline (FIG. 33) and .sup.1H NMR analysis showed peak shifting suggesting salt formation, with a likely 2:1 (acid:API) stoichiometry (data not shown). Residual THF (~0.5 molar eq.) also suggests a possible THF hemisolvate of Q203 salt.

(235) 5.11 Type 11 (1,5-Napthalenedisulfonic Acid)

(236) Type 11 material was generated from a slow evaporation experiment using NDSA and Q203 free base (2:1 acid/API) in a mixture of THF/acetone. XRPD analysis of Type 11 solid (ND-0006E-006-2.sub.4) showed the material was crystalline and the pattern was similar to Type 10 with minor differences (FIG. 34). .sup.1H NMR analysis showed peak shifting suggesting salt formation (data not shown). Residual THF (~0.7 molar eq.) also suggests a possible THF solvate. The stoichiometry of this salt has not been determined due to the presence of free acid.

(237) 5.12 Type 12 (Benzenesulfonic)

(238) Type 12 material was isolated from a sonication experiment using BSA and Q203 free base (1:1 acid/API) in THF. XRPD analysis of Type 12 solid (ND-0006E-005-15) showed the material was crystalline (FIG. 35). Proton NMR analysis showed peak shifting, ~0.03 molar eq. of THF suggesting salt formation with a 1:1 stoichiometry (data not shown).

(239) 5.13 Type 13 (Furoic)

(240) Type 13 material was generated from experiments involving furoic acid and Q203 free base (1:1 and 2:1 acid/API). Experimental conditions leading to Type 13 solid are highlighted in Table 15. XRPD analysis of Type 13 solid (ND-0006E-004-01) showed the material was crystalline (FIG. 36). .sup.1H NMR analysis in DMSO-d.sub.6 showed no peak shifting or residual solvent suggesting a possible polymorph of the free base, a Q203:furoic acid co-crystal or a degradant (data

not shown).

(241) TABLE-US-00024 TABLE 15 Experimental conditions leading to Type 13 solid Sample No. Anti- Acid:API (ND-0006E-) Solvent solvent Screen method (molar ratio) Result 004-01 MTBE none slurry (20° C.) 1:1 solid 005-01 THF none sonication 1:1 solid 007-01 THF methanol slow evap/slurry 2:1 solid (40° C.)

5.14 Type 14 (Citric)

(242) Type 14 material was generated from an ambient temperature slurring experiment using citric acid and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 14 solid (ND-0006E-004-03) showed the material was crystalline (FIG. 37) and .sup.1H NMR analysis showed no peak shifting or residual solvent suggesting a possible polymorph of the free base or a Q203: citric acid co-crystal (data not shown).

(243) 5.15 Type 15 (Fumaric)

(244) Type 15 material was isolated from slurring experiments using fumaric acid and Q203 free base (1:1 acid/API) in MTBE (20° C.) and IPA (40° C.). XRPD analysis of Type 15 solid (ND-0006E-004-04) showed the material was crystalline (FIG. 38) and .sup.1H NMR analysis showed no peak shifting or residual solvent suggesting a possible polymorph of the free base or a Q203: fumaric acid co-crystal (~0.7 molar eq. of fumaric acid) (data not shown).

(245) 5.16 Type 16 (Ketoglutaric)

(246) Type 16 material was generated from an ambient temperature slurring experiment using ketoglutaric acid and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 16 solid (ND-0006E-004-06) showed the material was crystalline (FIG. 39). Proton NMR analysis showed possibly peak shifting at 2.9 ppm, ~0.02 molar eq. of MTBE. Further analysis would need to be carried out in order to determine the nature of this Type (data not shown).

(247) 5.17 Type 17 (1,5-Napthalenedisulfonic Acid)

(248) Type 17 material was generated from an ambient temperature slurring experiment using NDSA and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 17 solid (ND-0006E-004-2.sub.4) showed the material was crystalline and the pattern was similar to Type 11 with both additional and missing peaks (FIG. 40). .sup.1H NMR analysis showed peak shifting with ~0.08 molar eq. of MTBE suggesting salt formation with likely a 2:1 acid/API stoichiometry (data not shown).

(249) 5.18 Type 18 (Maleic)

(250) Type 18 material was isolated from an ambient temperature slurring experiment using maleic acid and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 18 solid (ND-0006E-004-20) showed the material was crystalline and the pattern was similar to Type 19 with an additional peak present in the diffractogram (FIG. 41). .sup.1H NMR analysis showed peak shifting suggesting salt formation, with a likely 1:1 (acid:API) stoichiometry and residual MTBE (~0.2 eq) (data not shown).

(251) 5.19 Type 19 (Genticic)

(252) Type 19 material was generated from an ambient temperature slurring experiment using maleic acid and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 19 solid (ND-0006E-004-17) showed the material was crystalline and the pattern showed some similarity to Type 3 (FIG. 42). .sup.1H NMR analysis showed no peak shifting with some residual MTBE (0.1 molar eq.) and free acid (0.2 molar eq.) suggesting a possible polymorph of the free base or a Q203: gentisic acid co-crystal (data not shown).

(253) 5.20 Type 20 (p-Toluenesulfonic)

(254) Type 20 material was isolated from an ambient temperature slurring experiment using pTSA and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 20 solid (ND-0006E-004-16, FIG. 43) showed the material was crystalline and 1H NMR analysis showed peak shifting with trace amounts of residual MTBE (0.01 molar eq.) suggesting salt formation with a 1:1 acid/API stoichiometry (data not shown).

(255) 5.21 Type 21 (Tartaric)

(256) Type 21 material was isolated from slurrying experiments using tartaric acid and Q203 free base (1:1 and 2:1 acid/API) in MTBE and IPA (20° C. and 40° C. respectively). XRPD analysis of Type 21 solid (ND-0006E-004-1.sub.3) showed the material was crystalline (FIG. 44) and ¹H NMR analysis (data not shown) of solid (ND-0006E-007-13) showed no peak shifting or residual solvent with ~0.6 molar eq. of tartaric acid and an unknown peak at 8.1 ppm. TG/DTA analysis showed a weight loss of ~0.4% prior the melt observed at onset temperature of 168° C. These results suggests a possible co-crystal or polymorph of Q203 free base.

(257) 5.22 Type 22 (Succinic)

(258) Type 22 material was isolated from an ambient temperature slurrying experiment using succinic acid and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 20 solid (ND-0006E-004-12, FIG. 45) showed the material was crystalline and ¹H NMR analysis showed no peak shifting with trace amounts of residual MTBE (0.003 molar eq.) and ~1 molar eq. of succinic acid suggesting a possible co-crystal or polymorph of Q203 free base (data not shown).

(259) 5.23 Type 23 (Mandelic)

(260) Type 23 material was generated from experiments involving mandelic acid and Q203 free base (1:1 and 2:1 acid/API). Experimental conditions leading to Type 23 solid are highlighted in Table 16. XRPD analysis of Type 23 solid (ND-0006E-004-01) showed the material was crystalline (FIG. 46) with additional peaks observed in sample ND-0006E-007-09. ¹H NMR analysis of solid (ND-0006E-004-09) showed no peak shifting, trace amounts of residual solvent (0.01 molar eq.) and ~0.8 molar eq. of mandelic acid which may suggest possible polymorph or mandelic acid co-crystal of Q203 free base (data not shown).

(261) TABLE-US-00025 TABLE 16 Experimental conditions leading to Type 23 solid Sample No. Acid:API (ND-0006E-) Solvent Antisolvent Screen method (molar ratio) Result 004-09 MTBE none slurry (20° C.) 1:1 solid 005-09 THF none sonication 1:1 solid 007-09 IPA none slow evap/slurry 2:1 solid (40° C.)

5.24 Type 24 (Malic)

(262) Type 24 material was isolated from an ambient temperature slurrying experiment using malic acid and Q203 free base (1:1 acid/API) in MTBE and a sonication experiment using THF. XRPD analysis of Type 24 solid (ND-0006E-004-07) showed the material was disordered crystalline (FIG. 47) and ¹H NMR analysis of solid (ND-0006E-005-07) showed no peak shifting or residual solvent, with 1 molar eq. of malic acid which may suggest a possible polymorph or malic acid co-crystal of Q203 free base (data not shown).

(263) 5.25 Type 25

(264) Type 25 material was generated from sonication experiments involving various acids and Q203 free base (1:1 acid/API) in THF. Experimental conditions leading to pure Type 25 and Type 25 mixtures (with Type 3) are highlighted in Table 17. XRPD analysis of pure Type 25 solid (ND-0006E-005-13,20,27) showed the material was crystalline (FIG. 48). As this Type was obtained from different co-formers using THF, Type 25 is a likely a polymorph of Q203 free base and possibly solvated (THF).

(265) It should be noted that proton NMR analysis of Type 25 material (ND-0006E-005-20) showed peak shifting suggesting salt formation (1:1 acid/API) with ~0.5 molar eq. of THF which may indicate a possible hemi-solvate of the Q203 maleate salt (data not shown). This may be explained by the formation of a salt in-situ during preparation of the sample in deuterated methanol for NMR spectroscopy analysis.

(266) TABLE-US-00026 TABLE 17 Experimental conditions leading to Type 25 solid (pure and mixtures) Sample Anti- Acid:API Salt former No. (ND-0006E-) Solvent solvent (molar ratio) Result XRPD maleic 005-20 THF none sonication solid Type 25 saccharin 005-27 THF none sonication solid Type 25 tartaric (L) 005-13 THF none sonication solid Type 25 citric 005-03 THF none sonication solid Type 25 + Type 3 gentisic 005-17 THF none sonication solid Type 25

+ Type 3 HCl 005-18 THF none sonication solid Type 25 + Type 3 malonic 005-08 THF none sonication solid Type 25 + Type 3 oxalic 005-10 THF none sonication solid Type 25 + Type 3
phosphoric 005-26 THF none sonication solid Type 25 + Type 3 pyroglutamic 005-11 THF none sonication solid Type 25 + (L) Type 3 saccharin 003-27 THF acetone slow evap solution/solid Type 25 + Type 3

5.26 Type 26 (1,5-Naphthalenedisulfonic Acid)

(267) Type 26 material was generated from a slow evaporation followed by high temperature slurring experiment (40° C.) using NDSA and Q203 free base (2:1 acid/API) in IPA for 7 days. XRPD analysis of Type 26 solid (ND-0006E-007-24, FIG. 49) showed the material was crystalline and proton NMR spectroscopy shows peak shifting suggesting salt formation (data not shown) with ~0.04 molar eq. of IPA. The stoichiometry needs to be confirmed.

(268) 5.27 Type 27 (Gentisic)

(269) Type 27 solid was generated from a slow evaporation experiment using gentisic acid and Q203 free base (2:1 acid/API) in a mixture of THF/methanol. XRPD analysis of Type 27 solid (ND-0006E-006-17) showed the material was disordered crystalline (FIG. 50) and .sup.1H NMR analysis showed no peak shifting with ~0.5 molar eq. of THF, ~2 mol. eq. of co-former suggesting a possible hemisolvate of THF of gentisic co-crystal or polymorph of Q203 free base (data not shown).

(270) 5.28 Type 28 (Galactaric)

(271) Type 28 material was isolated from two LAG experiments using galactaric and gluconic acid and Q203 free base 1:1 acid:API in a mixture of IPA/water and acetone respectively. XRPD analysis of Type 28 (ND-0006E-008-15) showed the material was disordered crystalline (FIG. 51). Proton NMR analysis (data not shown) showed no peak shifting, ~0.07 molar eq. of IPA. TG/DTA results showed multiple endothermic events. Type 28 is likely a polymorph of the free base as obtained from different co-formers.

(272) 5.29 Type 29 (Ketoglutaric)

(273) Type 29 material was isolated from a sonication experiment using ketoglutaric acid and Q203 free base (1:1 acid/API) in THF. XRPD analysis of Type 29 solid (ND-0006E-005-06, FIG. 52) showed the material was disordered crystalline and similar to Type 25 with additional peaks observed. Proton NMR analysis showed no peak shifting, ~0.16 molar eq. of THF and ~0.9 molar eq. of ketoglutaric acid which may suggest a possible co-crystal or polymorph of Q203 free base (data not shown).

(274) 5.30 Type 30 (Fumaric)

(275) Type 30 material was isolated from a sonication experiment using ketoglutaric acid and Q203 free base (1:1 acid/API) in THF. XRPD analysis of Type 30 solid (ND-0006E-005-04, FIG. 53) showed the material was crystalline and .sup.1H NMR analysis (data not shown) showed no peak shifting or residual solvent, with ~0.85 molar eq. of fumaric acid which may suggest a possible polymorph or fumaric acid co-crystal of Q203 free base.

(276) 5.31 Type 31 (Sulphuric)

(277) Type 31 material was generated from a slow evaporation followed by high temperature slurring experiment (40° C.) using sulphuric acid and Q203 free base (2:1 acid/API) in IPA for 7 days. XRPD analysis of Type 31 solid (ND-0006E-007-28, FIG. 54) showed the material was crystalline and proton NMR spectroscopy showed significant peak shifting with no residual solvent suggesting salt formation (data not shown). It should be noted the presence of additional peaks which may be explained by degradation. The stoichiometry was not determined.

(278) 5.32 Type 32

(279) Type 32 material was generated from high temperature slurring experiments (40° C.) involving various acids and Q203 free base (2:1 acid/API) in IPA. Experimental conditions leading to pure Type 32 and Type 32 mixtures (with co-acid) are highlighted in Table 18. XRPD analysis of Type 32 solid (ND-0006E-007-06) showed the material was crystalline present (FIG. 55). .sup.1H

NMR analysis of Type 32 material showed no peak shifting (data not shown) and residual solvent and ~0.2 molar equivalent of co-former. As this Type was obtained from different co-formers, it is likely to be a polymorph of Q203 free base.

(280) TABLE-US-00027 TABLE 18 Experimental conditions leading to Type 32 solid (pure and mixtures) Sample No. Salt former (ND-0006E-) Solvent Result XRPD gentisic 007-17 IPA solid Type 32 gluconic (D) 007-25 IPA solid Type 32 glycolic 007-05 IPA solid Type 32 (hydroxyacetic) ketoglutaric 007-06 IPA solid Type 32 (oxoglutaric) maleic 007-20 IPA solid Type 32 malic (L) 007-07 IPA solid Type 32 malonic 007-08 IPA solid Type 32 phosphoric 007-26 IPA solid Type 32 pyroglutamic (L) 007-11 IPA solid Type 32 pyruvic (2- 007-21 IPA solid Type 32 oxopropanoic) succinic 007-12 IPA solid Type 32 galactaric (mucic) 007-29 IPA solid Type 32 + co-acid lactobionic 007-19 IPA solid Type 32 + additional peaks

5.33 Type 33 (Benzenesulfonic)

(281) Type 33 material was generated from a high temperature slurring experiment (40° C.) using BSA and Q203 free base (2:1 acid/API) in IPA for 7 days. XRPD analysis of Type 33 solid (ND-0006E-007-15, FIG. 56) showed the material was crystalline and proton NMR spectroscopy (data not shown) suggested possible salt formation with no residual solvent and likely a 2:1 acid/API stoichiometry. It should be noted similarities with Type 6 observed.

(282) 5.34 Type 34 (Ethane-1,2-Disulfonic Acid)

(283) Type 34 material was generated from a high temperature slurring experiment (40° C.) using EDSA and Q203 free base (2:1 acid/API) in IPA for 7 days. XRPD analysis of Type 34 solid (ND-0006E-007-22, FIG. 57) showed the material was disordered crystalline and proton NMR spectroscopy (data not shown) showed peak shifting and no residual solvent, suggesting salt formation with likely a 2:1 acid/API stoichiometry.

(284) 5.35 Type 35 (HCl)

(285) Type 35 material was generated from a slow evaporation followed by high temperature slurring experiment (40° C.) using HCl and Q203 free base (2:1 acid/API) in IPA for 7 days. XRPD analysis of Type 35 solid (ND-0006E-007-18, FIG. 58) showed the material was disordered crystalline and proton NMR spectroscopy (data not shown) showed peak shifting suggesting salt formation. No residual solvent or degradation was observed from the NMR spectrum. The stoichiometry has not been determined.

(286) 5.36 Type 36 (HCl)

(287) Type 36 material was isolated from an ambient temperature slurring experiment using HCl and Q203 free base (1:1 acid/API) in MTBE for 7 days. XRPD analysis of Type 36 solid (ND-0006E-004-18, FIG. 59) showed the material was crystalline, with similarities to the diffraction pattern of Type 3+additional peaks. Proton NMR analysis (data not shown) showed peak shifting with trace amounts of residual MTBE (0.007 molar eq.) and suggesting HCl salt of Q203. The stoichiometry has not been determined.

(288) 5.37 Type 37 (Phosphoric)

(289) Type 37 material was generated from high temperature slurring experiment (40° C.) and LAG technique using acetone and Q203 free base (1:1 acid/API). XRPD analysis of Type 37 solid (ND-0006E-008-13, FIG. 60) showed the material was crystalline. The slight baseline drift suggests the presence of amorphous content.

(290) Proton NMR analysis suggests peak shifting with around ~0.09 molar eq. of residual acetone Q203 phosphate salt (data not shown). The stoichiometry has not been determined.

(291) 5.38 Type 38 (Gluconic)

(292) Type 38 material was isolated from a co-melt experiment using gluconic and Q203 free base (1:1 acid/API). XRPD analysis of Type 38 solid (ND-0006E-009-07, FIG. 61) showed the material was crystalline. The baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed no peak shifting as expected, no trace of residual solvent and no presence of co-former. TG/DTA results (data not shown) showed a melt at onset temperature of

164° C. and another endotherm at onset 218° C. associated with weight loss. These results suggest Type 38 to be likely a polymorph of the free base.

(293) 5.39 Type 39 (Multiple Co-Formers)

(294) Type 39 material was isolated from co-melt experiments using multiple co-formers (malonic, pyruvic, saccharin) and Q203 free base (1:1 acid/API). XRPD analysis of Type 39 solid (ND-0006E-009-06, FIG. 62) showed the material was crystalline. A baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed no peak shifting with around 0.9 molar eq. of co-formers and suggesting potential co-crystal or polymorph of the free base.

(295) TABLE-US-00028 TABLE 19 Experimental conditions leading to Type 39 solid (pure and mixtures) Sample No. Acid:API Salt former (ND-0006E-) (molar ratio) Result XRPD malonic 009-03 1:1 molar eq. solid Type 39 pyruvic (2- 009-06 1:1 molar eq. solid Type 39 oxopropanoic) saccharin 009-09 1:1 molar eq. solid Type 39 + saccharin

5.40 Type 40 (Multiple Co-Formers)

(296) Type 40 material was isolated from HT slurry and LAG experiments using multiple co-formers (malonic, pyruvic, pyroglutamic, etc. . . .) and Q203 free base (1:1 acid/API). XRPD analysis of one of the Type 40 solid (ND-0006E-008-10, FIG. 63) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis showed no peak shifting with around 0.3 molar eq. of residual acetone and no trace of co-former suggesting likely polymorph of the free base (data not shown).

(297) TABLE-US-00029 TABLE 20 Experimental conditions leading to Type 40 solid (pure and mixtures) Sample No. Acid:API Salt former (ND-0006E-) (molar ratio) Result XRPD malonic 008-05 1:1 molar eq. solid Type 40 pyruvic (2- 008-10 1:1 molar eq. solid Type 40 oxopropanoic) glycolic 010-02 1:1 molar eq. solid Type 40 (hydroxyacetic) malonic 010-03 1:1 molar eq. solid Type 40 pyroglutamic (L) 010-06 1:1 molar eq. solid Type 40 pyruvic (2- 010-08 1:1 molar eq. solid Type 40 oxopropanoic) gluconic (D) 010-11 1:1 molar eq. solid Type 40 citric 008-03 1:1 molar eq. solid Type 40 + additional peaks glycolic 008-04 1:1 molar eq. solid Type 40 + (hydroxyacetic) additional peaks pyroglutamic (L) 008-07 1:1 molar eq. solid Type 40 + Form A + free acid

5.41 Type 41 (Multiple Co-Formers)

(298) Type 41 material was isolated from HT slurry and LAG experiments using multiple co-formers (lactobionic, ascorbic, 2-furoic, galactaric) and Q203 free base (1:1 acid/API).

(299) XRPD analysis of one of the Type 41 solid (ND-0006E-008-09, FIG. 64) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis showed no peak shifting with around 0.3 molar eq. of residual acetone and no trace of co-former suggesting likely Type 41 as a polymorph of the free base (data not shown).

(300) TABLE-US-00030 TABLE 21 Experimental conditions leading to Type 41 solid (pure and mixtures) Sample No. Acid:API Salt former (ND-0006E-) (molar ratio) Result XRPD lactobionic 010-01 1:1 molar eq. solid Type 41 ascorbic 010-07 1:1 molar eq. solid Type 41 lactobionic 008-01 1:1 molar eq. solid Type 41 2-furoic 010-14 1:1 molar eq. solid Type 41 + T28 galactaric (mucic) 010-01 1:1 molar eq. solid Type 41 + T28

5.42 Type 42 (Gentisic)

(301) Type 42 material was isolated from a LAG experiment using gentisic and Q203 free base (1:1 acid/API) in acetone. XRPD analysis of Type 42 solid (ND-0006E-008-08, FIG. 65) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis showed no peak shifting, ~0.14 molar eq. of acetone (~1.12% w/w) and ~0.67 molar eq. of co-former (data not shown). TG/DTA showed a weight loss from 25 to 113° C. (likely moisture) and an endotherm was observed at onset around 121° C. These results suggest either a polymorph or a potential co-crystal of Q203 free base. Further analysis would be needed to confirm one of these assumptions.

(302) 5.43 Type 43 (Ascorbic)

(303) Type 43 material was isolated from a LAG experiment using ascorbic and Q203 free base (1:1 acid/API) in a mixture of IPA/water. XRPD analysis of Type 43 solid (ND-0006E-008-02, FIG. 66) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed no peak shifting as expected, no residual IPA and ~0.8 molar eq. of co-former. TG/DTA showed a weight loss from 25 to ~131° C. of ~0.9% likely attributable to water moisture. Two endotherm events were observed at onset of 136 and 158° C. associated with a constant weight loss of ~11% from 130 to 300° C. These results suggest that Type 43 may be a mixture of polymorphs (free base and co-former) or a potential co-crystal of Q203. Further analysis would be required to better understand the nature of this Type.

(304) 5.44 Type 44 (Saccharin)

(305) Type 44 material was isolated from a LAG experiment using saccharin and Q203 free base (1:1 acid/API) in a mixture of IPA/water. XRPD analysis of Type 44 solid (ND-0006E-008-14, FIG. 67) showed the material was disordered crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed peak shifting with ~0.5 molar eq. of IPA suggesting IPA hemisolvate of Q203 saccharin salt. Further experiments may lead to unsolvated saccharin salt of Q203

(306) 5.45 Type 45 (Oxalic)

(307) Type 45 material was isolated from a HT slurry experiment (40 C) using oxalic and Q203 free base (1:1 acid/API) in acetone for 5 days. XRPD analysis of Type 45 solid (ND-0006E-010-04, FIG. 68) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis showed peak shifting with ~0.06 molar eq. of acetone suggesting salt formation of Q203 oxalate salt (data not shown).

(308) 5.46 Type 46 (Oxalic)

(309) Type 46 material was isolated from a HT slurry experiment (40 C) using oxalic and Q203 free base (1:1 acid/API) in methanol for 5 days. XRPD analysis of Type 46 solid (ND-0006E-010-05, FIG. 69) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed no peak shifting suggesting polymorph of the free base.

(310) 5.47 Type 47 (Saccharin)

(311) Type 47 material was isolated from a HT slurry experiment (40° C.) using saccharin and Q203 free base (1:1 acid/API) in a mixture of IPA/water for 5 days. XRPD analysis of Type 47 solid (ND-0006E-00-13, FIG. 70) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed peak shifting with trace of IPA, ~0.7 mol. eq. of co-former, suggesting salt formation of Q203 saccharin salt.

(312) 5.48 Type 48 (Salicylic)

(313) Type 48 material was isolated from a slow evaporation experiment using salicylic and Q203 free base (1:1 acid/API) in a mixture of THF/acetone. XRPD analysis of Type 47 solid (ND-0006E-003-34, FIG. 71) showed the material was crystalline. Proton NMR analysis showed no peak shifting, ~0.9 molar eq. co-former, ~0.1 molar eq. of THF and no residual acetone suggesting potential co-crystal or polymorph of the free base (data not shown).

(314) 5.49 Type 49 (nitric)

(315) Type 49 material was isolated from a sonication experiment using nitric and Q203 free base (1:1 acid/API) in THF. XRPD analysis of Type 49 solid (ND-0006E-005-30, FIG. 72) showed the material was crystalline, very similar to Type 3 with additional peaks. Proton NMR analysis (data not shown) showed peak shifting with ~0.25 molar eq. of THF suggesting likely Type 49 to be a nitrate salt of Q203.

(316) 5.50 Type 50 (Pamoic)

(317) Type 50 material was isolated from a RT slurry experiment using pamoic and Q203 free base (1:1 acid/API) in MTBE for 5 days. XRPD analysis of Type 50 solid (ND-0006E-004-33, FIG. 73)

showed the material was crystalline, very similar to Type 3 with additional peaks. Proton NMR analysis (data not shown) showed no peak shifting with ~0.07 molar eq. of MTBE and ~0.65 molar eq. of co-former suggesting possibly polymorph of Q203 free base or co-crystal.

(318) 5.51 Type 51 (Pamoic)

(319) Type 51 material was isolated from a sonication experiment using pamoic and Q203 free base (1:1 acid/API) in THF. XRPD analysis of Type 51 solid (ND-0006E-005-33, FIG. 74) showed the material was crystalline. Proton NMR analysis (data not shown) showed no peak shifting with ~0.02 molar eq. of THF and ~0.9 molar eq. of co-former suggesting possibly polymorph of Q203 free base or co-crystal.

(320) 5.52 Type 52 (Salicylic)

(321) Type 52 material was isolated from a RT slurry experiment using salicylic and Q203 free base (1:1 acid/API) in MTBE for 5 days. XRPD analysis of Type 52 solid (ND-0006E-005-33, FIG. 75) showed the material was crystalline and exhibiting similarities with Type 3. Proton NMR analysis (data not shown) is showed no peak shifting, ~0.36 molar eq. of co-former, ~0.40 molar eq. of MTBE suggesting likely a polymorph of Q203 free base.

(322) 5.53 Type 53 (Salicylic)

(323) Type 53 material was isolated from a sonication experiment using salicylic and Q203 free base (1:1 acid/API) in THF. XRPD analysis of Type 53 solid (ND-0006E-005-34, FIG. 76) showed the material was crystalline. Proton NMR analysis (data not shown) showed no peak shifting, ~1 molar eq. of co-former, ~0.04 molar eq. of THF (~0.4% w/w). TG/DTA analysis (data not shown) showed a weight loss of ~0.9% from 25° C. to ~125° C. likely attributable to moisture then a endotherm is observed at ~132° C. associated with weight loss up to 300° C. likely of ~20% which may be attributable to the loss of salicylic acid. These results suggest potential co-crystal of the Q203 FB but would need to be confirmed by further analysis.

(324) 5.54 Type 54 (Methanesulfonic)

(325) Type 54 material was isolated from an ambient slurry experiment using methanesulfonic acid and Q203 free base 1:1 acid:API in MTBE for 5 days. XRPD analysis of Type 54 (ND-0006E-004-23) showed the material was disordered crystalline and very similar to Type 3 (FIG. 77). Proton NMR analysis (data not shown) showed peak shifting, ~0.18 molar eq. of MTBE suggesting salt formation with 1:1 stoichiometry.

6 Assessment of Q203 Candidates (Salt/Co-Crystal)

(326) All the data generated from this study have been gathered in the Table below for selection of candidates of particular interest.

(327) TABLE-US-00031 TABLE 22 Assessment of Q203 salt/co-crystal solids (sort by co-former)
RH stress Aq. Sol Counter-ion (20 C./ Ease of NMR data TG/ (mg/ (class).sup.1 class Type
XRPD.sup.2 75 RH).sup.3 process (acid:API) DTA mL).sup.4 2-furoic Gras 13 ✓ ✓ — slurry (20°
C.) No PS, no — — sonication residual slow solvent evap/slurry (40° C.) ketoglutaric Gras 16 ✓
✓ ✓ — RT slurry PS at 2.9 ppm — — (oxoglutaric) or ✓ ✓ MTBE ~0.02 mol eq. MTBE 1:1 stoi.
ketoglutaric Gras 29 ✓ ✓ — sonication No PS, — — (oxoglutaric) or THF ~0.16 mol. eq. ✓ (very
THF lim qty) ~0.9 molr. eq. co-former saccharin Gras 44 ✓ — LAG PS, — — IPA/water Possibly
IPA hemi-solvate ~0.5 mol. eq. of IPA saccharin Gras 47 ✓ ✓ ✓ — HT slurry PS, — — (lim
IPA/water Trace of IPA, qty) ~0.7 mol. eq. of co-former urea Gras 5 ✓ ✓ ✓ — RT slurry No PS,
No — — MTBE and residual HT slurry IPA solvent Slow evap THF/MeOH ascorbic 1 43 ✓ ✓ —
LAG No PS, no — — or ✓ IPA/water residual solvent ~0.8 mol. eq. co-former citric 1 14 ✓ ✓ —
RT slurry No PS, no — — MTBE residual solvent fumaric 1 4 ✓ ✓ ✓ — slow evap no PS, ~0.5
— — MeOH/THF mol. eq. MeOH possibly hemisolvated of MeOH fumaric 1 15 ✓ ✓ ✓ —
Obtained 2 No PS, no — — times by residual Slurry (RT solvent and 40 C.) fumaric 1 30 ✓ ✓ —
Obtained No PS, no — — once, residual sonication solvent Galactaric 1 28 ✓ ✓ ✓ — LAG No
PS, — — or ✓ ✓ IPA/water ~0.07 mol. eq. of IPA- gluconic 1 38 ✓ ✓ — co-melt No PS, no — —

residual solvent HCl 1 35 ✓ ✓ — HT slurry IPA PS, no — (lim residual qty) solvent HCl 1 36 ✓ ✓ stable RT slurry PS, trace if — <0.31 (lim MTBE MTBE qty) maleic 1 18 ✓ ✓ ✓ — RT slurry PS, ~0.2 mol. — — or ✓ ✓ MTBE eq. MTBE, additional peaks 1:1 stoi. malic 1 24 ✓ ✓ — RT slurry No PS, no — — (L) (lim MTBE, residual qty) sonication solvent, 1 THF mol. eq. co-former phosphoric 1 37 ✓ ✓ ✓ — LAG acetone PS, ~0.09 — — mol. eq. acetone succinic 1 22 ✓ ✓ ✓ stable RT slurry No PS, no — <0.47 or ✓ ✓ MTBE residual (lim solvent qty) ~1 mol. eq. co-former sulphuric 1 31 ✓ ✓ ✓ Deliques- HT slurry IPA PS, no — <0.40 or ✓ ✓ cent residual (lim solvent, qty) additional peaks (deg) Tartaric 1 21 ✓ ✓ ✓ — RT and HT No PS, — — (L) or ✓ ✓ slurry using ~0.6 mol. eq. (lim MTBE/IPA co-former qty) One additional peak Benzene- 2 6 ✓ ✓ stable slow evap PS, ~0.25 — <0.54 sulfonic or ✓ THF/acetone mol. eq. of THF Benzene- 2 12 ✓ ✓ stable sonication PS, — <0.47 sulfonic or ✓ IPA ~0.03 mol. eq. of THF 1:1 stoi. Benzene- 2 33 ✓ ✓ ✓ — HR slurry IPA PS, no — — sulfonic residual solvent Likely 2:1 stoi. EDSA 2 9 ✓ ✓ stable slow evap PS, — <0.46 or ✓ THF/acetone ~0.07 mol. RT slurry eq. MTBE, MTBE 1:1 stoi sonication THF EDSA 2 34 ✓ ✓ — HT slurry IPA PS, no — — or ✓ residual solvent 2:1 stoi gentisic 2 19 ✓ ✓ ✓ — Obtained No PS, — — or ✓ ✓ once, slurry ~0.1 mol. eq. (RT) MTBE, ~0.2 mol. eq. co-former gentisic 2 27 ✓ ✓ — slow evap No PS, — — or ~0.5 mol. eq. ✓ (very THF, ~2 mol. lim qty) eq. co- former gentisic 2 42 ✓ ✓ ✓ — LAG No PS, — — or ✓ ✓ ~0.14 mol. eq. acetone ~0.7 mol. eq. co-former MSA 2 54 ✓ ✓ — RT slurry PS, — — (lim MTBE ~0.18 mol. qty) eq. of MTBE 1:1 stoi NDSA 2 10 ✓ ✓ or — Slow evap PS, — — ✓ THF/acetone, ~0.5 mol. eq. sonication THF THF 2:1 stoi NDSA 2 11 ✓ ✓ — Slow evap PS, — — THF ~0.7 mol. eq. THF Q203 salt NDSA 2 17 ✓ ✓ ✓ — RT slurry PS, — — or ✓ ✓ MTBE ~0.08 mol. eq. MTBE 2:1 stoi NDSA 2 26 ✓ ✓ ✓ stable HT slurry IPA PS, — <0.40 or ✓ ✓ ~0.04 mol. eq. IPA, oxalic 2 45 ✓ ✓ ✓ — HT slurry PS, — — acetone ~0.06 mol. eq. acetone pamoic 2 50 ✓ ✓ ✓ — RT slurry No PS, ~0.07 — — MTBE mol. eq. MTBE, ~0.65 mol. eq. co- former pamoic 2 51 ✓ ✓ ✓ — Sonication No PS, ~0.02 — — THF mol. eq. THF, ~0.9 mol. eq. co- former pTSA 2 Pat A ✓ ✓ ✓ stable HT slurry IPA Additional — <0.38 peak @3.45 ppm 2:1 stoi. pTSA 2 7 ✓ ✓ ✓ — Slow evap PS, no — — acetone, residual sonication solvent THF Additional peaks 1:1 stoi pTSA 2 8 ✓ ✓ ✓ — Slow evap PS, ~0.15 — — THF/acetone mol. eq. THF 2:1 stoi pTSA 2 20 ✓ ✓ ✓ stable RT slurry PS, ~0.01 — <0.30 MTBE mol. eq. MTBE 1:1 stoi mandelic 3 23 ✓ ✓ ✓ — RT and HT No PS, trace — — (DL) slurry MTBE residual and IPA, solvent, sonication ~0.8 mol. eq. THF co-former Nitric 3 49 ✓ ✓ — Sonication PS, — — or THF ~0.25 molar ✓ (very eq. THF lim qty) salicylic 3 48 ✓ ✓ ✓ — HT slurry No PS, ~0.1 — — THF/acetone mol. eq. THF, ~0.9 mol. eq. of co-former salicylic 3 52 ✓ ✓ — RT slurry No PS, ~0.4 — — MTBE mol. eq. MTBE ~0.4 mol. eq. co-former salicylic 3 53 ✓ ✓ ✓ — Sonication No PS, ~0.04 Melt — or ✓ ✓ THF mol. eq. THF onset at ~1 mol. eq. 132° C. co-former .sup.1Pharmaceutical class, class 1 = least toxic and unrestricted use as salt former, class 2 = low toxicity and good tolerability but not naturally occurring, class 3 = might be interested under particular circumstances .sup.2✓ ✓ ✓ = good crystallinity, = low crystallinity .sup.3Test for deliquescence, sample stressed for several days under 75% RH .sup.4Approximate values listed based of measured by aliquot addition on unbuffered water PS: peak shifting; stoi.: stoichiometry acid/API; mol. eq.: molar equivalent; RT and HT: room and high temperature

(328) A selection has been made from the previous results above based upon multiple criteria such as polymorphism landscape, crystallinity, class, nature of each Type (salt or potential co-crystal), specific toxicity, thermal behaviour, etc.

(329) The three candidates that the present inventors have identified as particularly useful are the first three entries in the following table (see Table 23).

(330) TABLE-US-00032 TABLE 23 Selection of Q203 salt/co-crystal solids (3 candidates shown in bold as first three entries) RH stress Aq. Sol Counter-ion (20 C./ Ease of NMR data TG/ (mg/

(class).sup.1 class Type XRPD.sup.2 75 RH).sup.3 process (acid:API) DTA mL).sup.4 HCl 1 36 ✓
 ✓ stable RT slurry PS, trace of — <0.31 (lim MTBE MTBE qty) phosphoric 1 37 ✓ ✓ ✓ —
 LAG PS, ~0.09 — — acetone mol. eq. acetone pTSA 2 20 ✓ ✓ ✓ stable RT slurry PS, ~0.01
 — <0.30 MTBE mol. eq. MTBE 1:1 stoi 2-furoic Gras 13 ✓ ✓ — slurry (20° C.) No PS, no —
 — sonication residual slow solvent evap/slurry (40° C.) ketoglutaric Gras 16 ✓ ✓ ✓ — RT slurry
 PS at 2.9 ppm — — (oxoglutaric) or ✓ ✓ MTBE ~0.02 mol. eq. MTBE 1:1 stoi. saccharin Gras 47
 ✓ ✓ ✓ — HT slurry PS, — — (lim IPA/water Trace of IPA, qty) ~0.7 mol. eq. of co-former urea
 Gras 5 ✓ ✓ ✓ — RT slurry No PS, No — — MTBE and residual HT slurry IPA solvent Slow
 evap THF/MeOH citric 1 14 ✓ ✓ — RT slurry No PS, no — — MTBE residual solvent fumaric 1
 15 ✓ ✓ ✓ — slurry No PS, no — — (RT and residual 40 C.) solvent fumaric 1 30 ✓ ✓ —
 Obtained No PS, no — — once, residual sonication solvent Galactaric 1 28 ✓ ✓ ✓ — LAG No
 PS, — — or ✓ ✓ IPA/water ~0.07 mol. eq. of IPA- gluconic 1 38 ✓ ✓ — co-melt No PS, no — —
 residual solvent HCl 1 35 ✓ ✓ — HT slurry IPA PS, no — — (lim residual qty) solvent maleic 1
 18 ✓ ✓ ✓ — RT slurry PS, ~0.2 mol. — — or ✓ ✓ MTBE eq. MTBE, additional peaks 1:1 stoi.
 malic 1 24 ✓ ✓ — RT slurry No PS, no — — (L) (lim MTBE, residual qty) sonication solvent, 1
 THF mol. eq. co- former phosphoric 1 37 ✓ ✓ ✓ — LAG acetone PS, ~0.09 — — mol. eq.
 acetone succinic 1 22 ✓ ✓ ✓ stable RT slurry No PS, no — <0.47 or ✓ ✓ MTBE residual (lim
 solvent qty) ~1 mol. eq. co-former Tartaric 1 21 ✓ ✓ ✓ — RT and HT No PS, — — (L) or ✓ ✓
 slurry using ~0.6 mol. eq. (lim MTBE/IPA co-former qty) One additional peak Benzene- 2 12 ✓ ✓
 stable sonication PS, — <0.47 sulfonic or ✓ IPA ~0.03 mol. eq. of THF 1:1 stoi. Benzene- 2 33 ✓
 ✓ ✓ — HR slurry IPA PS, no — — sulfonic residual solvent Likely 2:1 stoi. EDSA 2 9 ✓ ✓
 stable slow evap PS, — <0.46 or ✓ THF/acetone ~0.07 mol. RT slurry eq. MTBE, MTBE 1:1 stoi
 sonication THF EDSA 2 34 ✓ ✓ — HT slurry IPA PS, no — — or ✓ residual solvent 2:1 stoi
 gentisic 2 27 ✓ ✓ — slow evap No PS, — — or ~0.5 mol. eq. ✓ (very THF, ~2 mol. lim qty) eq.
 co- former gentisic 2 42 ✓ ✓ ✓ — LAG No PS, — — or ✓ ✓ ~0.14 mol. eq. acetone ~0.7 mol.
 eq. co-former MSA 2 54 ✓ ✓ — RT slurry PS, — — (lim MTBE ~0.18 mol. qty) eq. of MTBE 1:1
 stoi NDSA 2 17 ✓ ✓ ✓ — RT slurry PS, — — or ✓ ✓ MTBE ~0.08 mol. eq. MTBE 2:1 stoi
 NDSA 2 26 ✓ ✓ ✓ stable HT slurry IPA PS, — <0.40 or ✓ ✓ ~0.04 mol. eq. IPA 1:1 stoi oxalic 2
 45 ✓ ✓ ✓ — HT slurry PS, — — acetone ~0.06 mol. eq. acetone pamoic 2 51 ✓ ✓ ✓ —
 Sonication No PS, ~0.02 — — THF mol. eq. THF, ~0.9 mol. eq. co- former pTSA 2 Pat A ✓ ✓ ✓
 stable HT slurry IPA Additional — <0.38 peak @3.45 ppm 2:1 stoi. pTSA 2 20 ✓ ✓ ✓ stable RT
 slurry PS, ~0.01 — <0.30 MTBE mol. eq. MTBE 1:1 stoi mandelic 3 23 ✓ ✓ ✓ — RT and HT No
 PS, trace — — (DL) slurry MTBE residual and IPA, solvent, sonication ~0.8 mol. eq. THF co-
 former Nitric 3 49 ✓ ✓ — Sonication PS, — — or THF ~0.25 molar ✓ (very eq. THF lim qty)
 salicylic 3 48 ✓ ✓ ✓ — HT slurry No PS, ~0.1 — — THF/acetone mol. eq. THF, ~0.9 mol. eq. of
 co-former salicylic 3 52 ✓ ✓ — RT slurry No PS, ~0.4 — — MTBE mol. eq. MTBE ~0.4 mol. eq.
 co-former salicylic 3 53 ✓ ✓ ✓ — Sonication No PS, ~0.04 Melt — or ✓ ✓ THF mol. eq. THF
 onset ~1 mol. eq. at co-former 132° C. .sup.1Pharmaceutical class, class 1 = least toxic and
 unrestricted use as salt former, class 2 = low toxicity and good tolerability but not naturally
 occurring, class 3 = might be interested under particular circumstances .sup.2✓ ✓ ✓ = good
 crystallinity, = low crystallinity 3Test for deliquescence, sample stressed for several days under
 75% RH .sup.4Approximate values listed based of measured by aliquot addition on unbuffered
 water PS: peak shifting; stoi.: stoichiometry acid/API; mol. eq.: molar equivalent; RT and HT:
 room and high temperature

7 Scale Up of Selected Salts

(331) 7.1 P-TSA (Nd-6E-012-01)

(332) API (1 g) was added to a scintillation vial containing THF (18 ml). Seeds of Type 20 solid (ND-0007E-004-16) were added to the solution, which did not dissolve. pTSA dissolved in acetone

(342 mg, 18 ml) was added to the API solution. Precipitation was observed with stirring. Proton NMR analysis after T=1d suggested the pTSA salt had formed (1.3 molar eq. with additional peak at 8.6 ppm). The solid was isolated and slurried in MTBE for 3 days. XRPD analysis of the resultant solid confirmed Type 20.

(333) 7.2 Phosphate (ND-6E-012-02)

(334) Concentrated H.sub.2PO.sub.4 (122.8 ul) was added to a scintillation vial containing acetone (7.5 ml). Seeds of Type 37 solid (ND-0007E-008-13) were added to the solution, which did not dissolve. API (1 g) was added to the vial with stirring resulting in a block of solid. This was vortexed until a thick slurry was obtained. The slurry (not well mixable) became more mixable after ~10 mins at 40° C. A pull was taken at T=1 day for proton NMR analysis which showed ~0.43 molar eq. of acetone. The slurry was left for T=5 days, prior to isolation of the solid by filtration. XRPD analysis showed Type 37.

(335) 7.3 HCl (ND-6E-012-03)

(336) Concentrated HCl (147.3 ul) was added to a scintillation vial containing THF (18 ml). Seeds of Type 36 solid (ND-0007E-004-18) were added to the solution, which dissolved. The HCl/THF solution was added to API (1 g) resulting in a solution. The solution was again seeded with T36 solid, which dissolved. The solution was subjected to evaporation under nitrogen for ~5 mins until the solution became a haze. Seeds of T36 were added resulting in a suspension. The solution was evaporated under nitrogen with stirring. XRPD showed a novel Type 55. Proton NMR analysis showed a peak shifting confirming salt formation.

8 pH Profile Analysis

(337) The pH profile has been assessed on the three salt candidates of Q203 (monotosylate, phosphate and HCl). The HPLC method is detailed in Section 2.3.6.

(338) 8.1 HPLC Method Check

(339) The HPLC method was initially checked for suitability of use, and a working range from 0.0625 mg/mL to 0.5 mg/mL was established for Q203. (Note diluent was changed to methanol, as material was not soluble in diluent listed in method supplied). Samples from solubility experiments were diluted where required so the API concentration fell within the working concentration range. A linearity of R²=0.999 was found showing a really good fit (data not shown).

(340) 8.2 pH Solubility Profiling

(341) Solubility of salts at pH 1, 4.5, 6.8 and 7.5 was determined as described in section 2.3.6 and 2.3.7. pTSA remained the same pattern following slurrying in pH 1 buffer and exhibited an average solubility of ~0.33 mg/mL. However at pH 4.5 and above the pTSA salt converted to Type 28 which was found to be a polymorph of the freebase from XRPD and proton NMR analyses and had shown a very limited solubility (<0.00037 mg/ml), see table 25.

(342) This was also the case for the phosphate salt at pH 4.5 and above, but following slurrying in pH 1 buffer converted to a new HCl salt, Type 56. Type 56 was also observed from all solids isolated from HCl solubility experiments. The greatest solubility was ~5.58 mg/mL observed from HCl salt Type 56 at pH 1 (sample LMcG-0007E-004-09_1 was discounted as this figure was significantly lower than other three values, and may have been a dilution error). Selected results (XRPD traces of solids isolated after pH solubility experiments) are also shown in FIG. 78.

(343) TABLE-US-00033 TABLE 24 Results of pH solubility profiling Sample no HPLC (LMcG-Pull concentration 0007E- Final time based on avg 004-) Input pH (hrs) XRPD NMR std (mg/ml)

— Ditosylate 1	— — — 1.45 (client result) 01_1	pTSA salt, 1.2	24	p-TSA salt — 0.33	Type 20
T20 (ND-0006E- crystalline 012-01)	01_2	pTSA salt, 1.14	24	p-TSA salt — 0.32	Type 20
T20 (ND-0006E- crystalline 012-01)	02_1	pTSA salt, 4.1	23	Type 28 — 3.66E-04	Type 20
disordered (ND-0006E- 012-01)	02_2	pTSA salt, 4.47	23	Type 28 — 3.33E-04	Type 20
disordered (ND-0006E- 012-01)	03_1	pTSA salt, 6.45	22.5	Type 28 no PS 3.12E-04	Type 20
disordered (ND-0006E- 012-01)	03_2	pTSA salt, 6.71	22.5	Type 28 — 3.07E-04	Type 20
disordered (ND-0006E- 012-01)	04_1	pTSA salt, 7.49	72	Type 28 — 1.20E-04	Type 20

disordered (ND-0006E- 012-01) 04_2 pTSA salt, 7.3 72 Type 28 — 3.97E-05 Type 20
disordered (ND-0006E- 012-01) 05_1 Phosphate 1.23 24 T56 HCl — 5.33 salt, Type 37 salt (ND-0006E- v. 012-02) disordered 05_2 Phosphate 1.18 24 T56 HCl higher 5.64 salt, Type 37 salt PS
(ND-0006E- v. 012-02) disordered 06_1 Phosphate 4.1 23 Type 28 — 3.84E-04 salt, Type 37
disordered (ND-0006E- 012-02) 06_2 Phosphate 4.27 23 Type 28 — 3.36E-04 salt, Type 37
disordered (ND-0006E- 012-02) 07_1 Phosphate 6.33 22.5 Type 28 — 2.80E-04 salt, Type 37
disordered (ND-0006E- 012-02) 07_2 Phosphate 6.79 22.5 Type 28 — 2.97E-04 salt, Type 37
disordered (ND-0006E- 012-02) 08_1 Phosphate 7.54 72 Type 28 — 6.39E-05 salt, Type 37
disordered (ND-0006E- 012-02) 08_2 Phosphate 7.41 72 Type 28 — 4.10E-05 salt, Type 37
disordered (ND-0006E- 012-02) 09_1 HCl salt, 1.16 24 T56 HCl PS 2.54.sup.1 Type 36 salt
(ND-0006E- crystalline 012-03) 09_2 HCl salt, 1.09 24 T56 HCl slight std (mg/ml) Type 36 salt
PS 5.76 (ND-0006E- crystalline 012-03) 10_1 HCl salt, 4.34 23 T56 HCl slight 3.23E-04 Type
36 salt PS (ND-0006E- crystalline 012-03) 10_2 HCl salt, 4.34 23 T56 HCl — 3.07E-04 Type 36
salt (ND-0006E- crystalline 012-03) 11_1 HCl salt, 6.99 22.5 T56 HCl — 2.85E-04 Type 36 salt
(ND-0006E- crystalline 012-03) 11_2 HCl salt, 6.9 22.5 T56 HCl — 2.77E-04 Type 36 salt (ND-0006E- crystalline 012-03) 12_1 HCl salt, 7.28 72 T56 HCl — 3.28E-05 Type 36 salt (ND-0006E- crystalline 012-03) 12_2 HCl salt, 7.28 72 T56 HCl — 3.57E-05 Type 36 salt (ND-0006E- crystalline 012-03) .sup.1possible dilution error

8.3 Conclusions of the pH Profile Analysis

(344) pTSA salt remained stable at pH 1 buffer and had a solubility of ~0.33 mg/mL, however at pH 4.5 and above converted to freebase (type 28) which had lower solubility. Phosphate salt was not stable in any pH buffer, at pH 1 converted to HCl Type 56 and at pH 4.5 and above converted to freebase. Type 36 HCl salt converted to new type HCl salt Type 56 in all buffers tested.

(345) Improvement of the solubility was found for the three candidates compared to the free base and the current candidate (ditosylate salt). It should be noted that below pH 4.5, phosphate and pTSA exhibited a conversion to the free base suggesting some degree of instability.

(346) On the other hand, HCl salt exhibited a higher solubility and under the conditions tested no conversion to the free base was observed. This may be explained by low wettability, kinetic factors (time, stirring effect).

9 Conclusion

(347) A salt screen was performed on Q203 using thirty seven salt formers in various solvent systems. More than 50 new Types were observed which showed a very high tendency of Q203 to generate new patterns (polymorph of the free base but mainly salts and potential co-crystals).

(348) These new patterns have been analysed by different analytical techniques (eg. XRPD, proton NMR, aqueous solubility when sufficient material was available).

(349) A selection of three salt candidates for further pH profile analysis has been made upon multiple criteria such as polymorphism landscape, crystallinity, class, nature of each Type (salt or potential co-crystal), specific toxicity, thermal behaviour, etc. All potential co-crystals have not been considered as further characterisation would be needed to confirm the nature of these solids (whether or not co-crystal). Pharmaceutical class of the counter ion has also been taken into account (HCl and phosphoric acid are considered as Class 1 which are defined as least toxic and unrestricted use as salt former. The monotosylate salt has been chosen although it is considered as Class 2 (low toxicity and good tolerability but not naturally occurring). By comparison between mono and ditosylate, the monotosylate would exhibit a lower toxicity as only one molecule of counter ion per molecule of API will be present opposed as the ditosylate salt.

(350) It should be noted that other candidates may be of interest for further development but further investigation would be needed and compared to these 3 candidates.

(351) Improvement of the solubility was found for the three candidates compared to the free base and the ditosylate salt. It should be noted that below pH 4.5, phosphate and pTSA exhibited a conversion to the free base suggesting some instability.

(352) On the other hand, HCl salt exhibited the higher solubility and under the conditions tested no conversion to the free base was observed. This may be explained by low wettability, kinetic factors (time, stirring effect) especially above pH4.

APPENDIX 1

(353) TABLE-US-00034 TABLE 25 Summary of the observed solids of Q203 (sorted out by co-former) Comments/ Type Co-former Tentative Assignments 13 2-furoic Crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests no salt formation Potential co-crystal or polymorph of Q203 free base 43 ascorbic Disordered crystalline, prepared by LAG using a mixture of IPA/water Proton NMR: no PS, ~0.8 molar eq. co-former, no residual IPA 6 benzenesulfonic Disordered crystalline, prepared by slow evaporation using THF/acetone .sup.1H NMR suggests salt formation with ~0.25 molar eq. of THF Q203 benzenesulfonate salt (1:1 or 2:1 stoichiometry) 12 benzenesulfonic Crystalline, prepared by sonication using THE .sup.1H NMR suggests salt formation with ~0.03 molar eq. of THF Q203 benzenesulfonate (1:1 stoichiometry) 33 benzenesulfonic Crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with no residual solvent Q203 benzenesulfonate salt (likely 2:1 stoichiometry) 14 citric Crystalline, prepared by RT slurring for 7 d using MTBE .sup.1H NMR suggests no salt formation Potential co-crystal or polymorph of Q203 free base 9 ethane-1,2-disulfonic Disordered crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests salt formation with ~0.07 molar eq. of MTBE Q203 edisylate (potential 1:1 stoichiometry) 34 ethane-1,2-disulfonic Disordered crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with no residual solvent Q203 edisylate salt (2:1 stoichiometry) 4 fumaric Crystalline, prepared by slow evaporation using a mixture of THF/MeOH .sup.1H NMR suggests no salt formation and ~0.5 eq residual MeOH Potential MeOH hemi-solvate of co-crystal or polymorph of Q203 free base 15 fumaric Crystalline, prepared by RT/HT slurring for 7 d using MTBE/IPA .sup.1H NMR suggests no salt formation (~0.7 molar eq. fumaric acid) Potential co-crystal or polymorph of Q203 free base 30 fumaric Crystalline, prepared by sonication using THF .sup.1H NMR shows no peak shifting, no residual solvent and ~0.9 molar eq. of fumaric acid suggesting possible co-crystal or polymorph of Q203 free base. 28 galactaric, gluconic Crystalline, prepared by LAG experiments using a mixture of IPA/water and acetone from galactaric and gluconic respectively .sup.1H NMR suggests no salt formation (~0.08 molar eq. IPA). Further obtained as part of the pH profile analysis so likely polymorph of the Free base 19 gentisic Crystalline (similar to T3), prepared by RT slurring for 7 d using MTBE .sup.1H NMR suggests no salt formation (~0.1 eq residual MTBE and 0.2 eq. free acid) Potential co-crystal or polymorph of Q203 free base 27 gentisic Disordered crystalline, prepared by slow evaporation using a mixture THF/methanol .sup.1H NMR shows no peak shifting, with ~0.5 molar eq. of THF, ~2 mol. eq. of co-former, suggesting possible THF hemisolvate of co-crystal or polymorph of the free base. 42 gentisic Disordered crystalline, prepared by LAG using a mixture of IPA/water 38 gluconic Crystalline, prepared by co-melt Proton NMR: no PS as expected, no solvent, no presence of co-former TG/DTA: melt at ~165° C. Likely polymorph of the free base 35 HCl Disordered crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with no residual solvent Q203 HCl salt. The stoichiometry has not been determined 36 HCl Crystalline, prepared by HT slurring for 7 d using IPA .sup.1H NMR suggests salt formation with ~0.007 molar eq. of MTBE Q203 HCl salt. The stoichiometry has not been determined 55 HCl Crystalline with the presence of disordered material, prepared by evaporation from THE Proton NMR: shifting confirming salt formation. 56 HCl Crystalline with the presence of disordered material, generated as part of the pH profile analysis from Type 55 and phosphate salt at pH1 Proton NMR: shifting confirming salt formation 16 ketoglutaric Crystalline, prepared by RT slurring for 7 d using MTBE (oxoglutaric) .sup.1H NMR shows possibly PS at 2.9 ppm, 0.02 eq residual MTBE. Nature of this Type needs to be confirmed 29 ketoglutaric Disordered crystalline, prepared by sonication using THE (oxoglutaric) .sup.1H NMR shows no peak shifting, with ~0.16 molar eq. of THF and

~0.9 molar eq. of ketoglutaric acid suggesting possible co-crystal or polymorph of Q203 free base. 18 maleic Crystalline (similar to T19), prepared by RT slurring for 7 d using MTBE .sup.1H NMR suggests salt formation and ~0.2 eq residual MTBE Potential Q203 maleate (1:1 stoichiometry), possible non-stoichiometric MTBE solvate 24 malic (L) Crystalline, prepared by RT slurring using MTBE and by sonication using THF .sup.1H NMR shows no peak shifting, no residual solvent and ~1 molar eq. of malic acid suggesting possible co-crystal or polymorph of the free base. 23 mandelic (DL) Crystalline, prepared by RT and HT slurring using MTBE and IPA, and by sonication 1H NMR shows no Peak shifting and ~0.8 molar eq. of mandelic acid suggesting possible co-crystal or polymorph of the free base. 54 methanesulfonic Disordered crystalline, prepared by RT slurry for 7 d using MTBE Very similar to Type 3 Proton NMR: peak shifting with ~0.18 molar eq. of MTBE suggesting likely MSA salt of Q203 with a 1:1 stoichiometry. 2 multiple co-formers Crystalline, prepared by slow evaporation using THE/MeOH .sup.1H NMR suggests no salt formation and no visible degradation Potential polymorph of Q203 free base 3 multiple co-formers Crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests no salt formation. Potential polymorph of Q203 free base 25 multiple co-formers Crystalline, prepared by multiple solvents and techniques obtained as pure or as a mixture with Type 3. This suggests polymorph of the free base. But .sup.1H NMR analysis of material from maleic suggests salt formation with residual solvent (~0.5 molar eq THF). Likely a THF hemi-solvate of the salt of Q203. This may be explained by possible salt formation in-situ during NMR preparation sample. 32 multiple co-formers Crystalline, prepared by HT slurring for 7 d using IPA and stoichiometry .sup.1H NMR shows no peak shifting, no residual solvent and ~0.2 molar eq. of ketoglutaric acid. As obtained from multiple co-formers, likely a polymorph of Q203 free base. 39 multiple co-formers Crystalline, prepared from various co-formers (malonic, pyruvic and saccharin) by co-melting technique suggesting polymorph of the free base Proton NMR of saccharin and pyruvic solids showed salt formation suggesting salt formation had occurred during NMR sample preparation. 40 multiple co-formers Crystalline, prepared from various co-formers (gluconic, malonic, pyruvic. . .) and various techniques co-melting technique suggesting polymorph of the free base 41 multiple co-formers Crystalline, prepared from various co-formers (ascorbic, lactobionic, galactaric) and techniques (HT slurry, LAG) Likely polymorph of the free base 1 n/a material, crystalline, free based .sup.1H NMR analysis showed no visible degradation and trace of residual solvent (may be acetone around ~80 ppm). 10 naphtalene-1,5- Crystalline, prepared by a range of techniques and solvents disulfonic .sup.1H NMR suggests salt formation and ~0.5 molar eq. residual THF Potential THF hemisolvate of Q203 naphthalenedisulfonate (2:1 stoichiometry) 11 naphtalene-1,5- Crystalline (similar to T10), prepared by slow evaporation using THF disulfonic .sup.1H NMR suggests salt formation and ~0.7 molar eq. residual THF Potential THF solvate of Q203 naphthalenedisulfonate Stoichiometry not determined due to the presence of free acid 17 naphtalene-1,5- Crystalline (similar to T11), prepared by RT slurring for 7 d using MTBE disulfonic .sup.1H NMR suggests salt formation with ~0.08 molar eq. of MTBE Potential Q203 NDSA salt (2:1 stoichiometry) 26 naphtalene-1,5- Crystalline, prepared by slow evap, followed by HT slurring for 7 d using disulfonic IPA .sup.1H NMR suggests salt formation (~0.04 molar eq. IPA). The stoichiometry needs to be confirmed 49 nitric Disordered crystalline, prepared by sonication using THF Very similar to Type 3 + additional peaks Proton NMR: peak shifting with ~0.25 molar eq. of THF suggesting likely Type 49 to be a nitrate salt of Q203. 45 oxalic Crystalline, prepared by HT slurry using acetone Proton NMR: peak shifting with ~0.06 molar eq. of acetone. Stoichiometry not determined. Oxalate salt of Q203. 46 oxalic Crystalline, prepared by HT slurry using methanol Proton NMR: no peak shifting. Stoichiometry not determined. Likely polymorph of the free base. 50 pamoic Crystalline, prepared by RT slurry using MTBE Similarities with Type 3 Proton NMR: no peak shifting with ~0.07 molar eq. of MTBE and ~0.65 molar eq. of co-former Possibly polymorph of Q203 free base or co-crystal. 51 pamoic Crystalline, prepared by sonication using THF Proton NMR: no PS, ~0.9 molar eq. co-former, ~0.02 molar eq. of THF

Potential polymorph or co-crystal of Q203. 37 phosphoric Crystalline, prepared by HT slurry and by LAG using acetone Proton NMR showed peak shifting with ~0.09 molar eq. of acetone Suggesting Q203 phosphate salt. The stoichiometry has not been determined Pat p-toluenesulfonic Crystalline, prepared by slow evap. then HT slurrying using IPA, XRPD A pattern similar to the ditosylate salt (Pattern A) .sup.1H NMR confirmed salt formation with a 2:1 stoichiometry 7 p-toluenesulfonic Crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests salt formation with no residual solvents. Additional peaks were observed. Q203 tosylate (potential 1:1 stoichiometry) 8 p-toluenesulfonic Crystalline, prepared by slow evaporation using THF/acetone .sup.1H NMR suggests salt formation with ~0.15 molar eq. of THF Q203 tosylate (potential 2:1 stoichiometry) 20 p-toluenesulfonic Crystalline, prepared by RT slurrying for 7 d using MTBE .sup.1H NMR suggests salt formation (~0.01 molar eq. MTBE) Q203 tosylate (1:1 stoichiometry) 44 saccharin Disordered material, very similar to Type 41, prepared by LAG using a mixture of IPA/water. Proton NMR: peak shifting, ~1 molar eq. of saccharin and ~0.5 molar eq. of IPA suggesting IPA hemisolvate of saccharin salt 47 saccharin Crystalline, prepared by HT slurry using a mixture of IPA/water Proton NMR: peak shifting with trace of IPA, ~0.7 mol. eq. of co-former suggesting salt formation of Q203 saccharin salt. 48 salicylic Crystalline, prepared by HT slurry using a mixture of THF/acetone Proton NMR: no peak shifting, ~0.9 molar eq. co-former, ~0.1 molar eq. of THF and no residual acetone suggesting potential co-crystal or polymorph of the free base. 52 salicylic Crystalline, prepared by RT slurry using MTBE Similarities with Type 3 Proton NMR: no PS, ~0.4 molar eq. co-former, ~0.4 molar eq. of MTBE Likely polymorph of Q203. 53 salicylic Crystalline, prepared by sonication using THF Proton NMR: no PS, ~1.0 molar eq. co-former, ~0.04 molar eq. of THF TG/DTA: melt at ~132° C. Potential co-crystal of Q203. 22 succinic Crystalline, prepared by RT slurrying for 7 d using MTBE .sup.1H NMR suggests salt formation (~0.01 molar eq. MTBE) Potential Q203 tosylate (1:1 stoichiometry) 31 sulphuric Crystalline, prepared by HT slurrying for 7 d using IPA .sup.1H NMR suggests salt formation with no residual solvent and the presence of additional peaks (possibly degradation). The stoichiometry has not been determined 21 tartaric (L) Crystalline, prepared by RT/HT slurrying for 7 d using MTBE/IPA .sup.1H NMR suggests no salt formation (~0.6eq tartaric acid + unknown peak at 8.1 ppm) Potential co-crystal or polymorph of Q203 free base 5 urea Crystalline, prepared by a range of techniques and solvents .sup.1H NMR suggests no salt formation, no residual solvent Potential co-crystal or polymorph of Q203 free base

Example 3

(354) 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide (Q203)-ditosylate form A was subjected to stability experiments. More specifically, samples of Q203-ditosylate form A were exposed to an environment of 60% RH (relative humidity) and 25° C. for periods of time between 6 months and 60 months. Furthermore, further stability experiments were done with some samples of Q203-ditosylate form A under accelerated conditions (40° C.-75% relative humidity (RH)). These experiments under accelerated conditions were done for a period of up to 6 months. Subsequently, samples were analyzed by HPLC and checked for impurities. It turned out that under conditions of 25° C. and 60% RH, Q203-ditosylate form A remains stable and does not decompose or otherwise deteriorates and there were only minor impurities that could be identified in the corresponding HPLC-chromatograms (chromatogram traces not shown). The results are summarized in the following table:

(355) TABLE-US-00035 TABLE 27 Stability results for form A of Q203-ditosylate at 25° C./60% RH Purity [%] Time [months] (as measured by HPLC) 0 >98% 6 >98% 12 >98% 24 >98% 36 >98% 48 >98% 60 >98%

(356) It can be seen that the purity remains >98% and does not show any sign of decomposition or deterioration.

(357) Furthermore, in the experiments under accelerated conditions, there was no decomposition for a period up until 6 months when the sample still showed a purity of >98% (data not shown).

(358) From these data it can be concluded that Q203-ditosylate form A is stable under long-term storage conditions of 25° C. and 60% RH for up to 60 months and under accelerated conditions of 40° C. and 75% RH up to 6 months.

Claims

1. Compound 6-chloro-2-ethyl-N-(4-(4-(4 (trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide ditosylate having the structure ##STR00003## having an XRPD spectrum essentially as shown in FIG. 1.
2. The compound according to claim 1, having a differential scanning calorimetry (DSC) thermogram showing a single endotherm peak with an onset of 235° C.-237° C.
3. The compound according claim 1, being produced by a method comprising the steps: providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2; mixing and dissolving them in a suitable solvent or solvent mixture; and evaporating the solvent or solvent mixture.
4. The compound according to claim 3, wherein the solvent or solvent mixture is isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THE and acetone.
5. A mono-acid addition salt of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide which is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-hydrochloride, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-phosphate, or 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-tosylate; which mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-hydrochloride having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 6.4° 2θ, 8.1° 2θ, 16.2° 2θ, 17.2° 2θ, 24.3° 2θ and 25.0° 2θ, +0.2° 2θ; or which mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-phosphate having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 9.0° 2θ, 10.7±0.2° 2θ, 11.7° 2θ, 14.8° 2θ, 18.4° 2θ, 19.3° 2θ, and 21.8° 2θ, 22.8° 2θ, 0.2° 2θ; or which mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-tosylate having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K.sub.α-radiation (Cu-K.sub.α): 4.0° 2θ, 11.4° 2θ, 12.2° 2θ, 14.4° 2θ, 17.7° 2θ, 18.9° 2θ, 19.7° 2θ, 20.3° 2θ, 23.2° 2θ, and 26.7° 2θ, ±0.2° 2θ.
6. The mono-hydrochloride salt according to claim 5, having an XRPD spectrum essentially as shown in FIG. 2a.
7. The mono-phosphate salt according to claim 5, having an XRPD spectrum essentially as shown in FIG. 2b.
8. The mono-tosylate salt according to claim 5, having an XRPD spectrum essentially as shown in FIG. 3.
9. A pharmaceutical composition comprising at least one compound according claim 1 or a mono-acid addition salt according to claim 5, together with at least one pharmaceutically acceptable carrier, excipient and/or diluent.
10. The pharmaceutical composition according to claim 9, further comprising at least one other pharmaceutically active agent.
11. A method for making the compound of claim 1, said method comprising the steps: providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-

yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2; mixing and dissolving them in a suitable solvent or solvent mixture; and evaporating the solvent or solvent mixture.

12. The method according to claim 11, wherein the solvent or solvent mixture is isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THE and acetone.

13. A method for preparing the mono-acid addition salt of claim 5, said method comprising the steps: providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and an acid selected from hydrochloric acid, phosphoric acid and para-toluenesulphonic acid in a stoichiometric ratio of 1:1; mixing and dissolving them in a suitable solvent or solvent mixture; and evaporating the solvent or solvent mixture.

14. The method according to claim 13, wherein the solvent or solvent mixture is isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THE and acetone.

15. A method for the treatment of a bacterial infection, comprising application of a suitable amount of a compound according to claim 1 or claim 5 to a patient in need of such treatment.

16. The method of claim 15, wherein the bacterial infection is tuberculosis or Buruli ulcer.
